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英文论文题目： **From Narrative to Conversion: The Dynamic Impact of Streamer Storytelling on Livestream Sales Performance — An Empirical Study of Taobao Women's Apparel**

申请人姓名：KOOI LI YNG

指导教师：陈熹

专业名称：管理科学与工程

研究方向：社交媒体

所在学院：管理学院

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论文作者签名: _____

指导教师签名: _____

论文评阅人 1：_____

评阅人 2：_____

评阅人 3：_____

答辩委员会主席：瞿文光 副教授 博导 浙江大学管理学院

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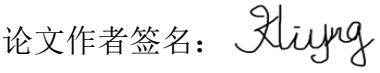
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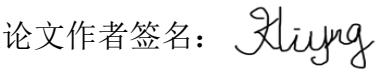
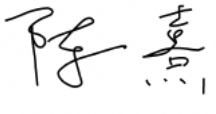
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致 谢

本以为“致谢”会是整篇论文中最让人期待、最能自由发挥的部分，可当真的落笔于此时，竟然有些不知从何说起。

硕士的这三年时光，看似漫长却又转瞬即逝。对于我这名留学生而言，这三年无疑是我人生中学习与成长曲线（learning curve）最为陡峭，也是收获最丰盛的一段岁月。回想起当初决定赴华就读时，亲朋好友不禁担忧这会不会又是一次“青春年华的赌注”。毕竟在亚洲的固有社会观念中，年龄常常被设定为衡量人生阶段与目标的刻度。但对我而言，人生既短暂又充满无限的可能与选择。我愿选择走一条充满未知的窄路，也不愿错过任何尝试的机会，哪怕前方充斥着困难与失败的风险。社会灌输给我们的成功秘诀，往往是“要在对的时间点做对的选择”。但我想，何谓真正的“对的时间点”与“对的选择”？人，正因为对现时感到茫然，对未来心存未知，才更能体会活在盼望之中、乐在成就之内的意义。成功也许可以复刻，但伟大的人生却不然。每个人都是独特的，都该自行书写并描绘属于自己的“人生叙事”。至于我，我只想拥有一段“有趣”的人生——无论处于什么年龄，都保持对探索生命未知的热情。正如我当初决定来中国开启留学生活，以及此刻，我终于跨越了自以为的软弱，完成了这篇曾以为自己根本无法胜任的论文。

在此，我要衷心感谢我的导师陈熹教授。感谢您在论文撰写过程中耐心帮助我构思框架、理顺逻辑；也要感谢瞿文光教授在关键时刻的严格把关与指点；感谢朱泉老师对论文规范格式提出的宝贵意见；以及感谢田雨老师，总是对我这名留学生充满耐心，细致地解答我在课程与教务上的各种问题。

感谢我的父母，是你们永远无条件地支持我每一次看似“离经叛道”的选择。还有我最好的朋友西西，我们的相遇虽在大西洋的彼岸，但正是因为有你的

鼓励与我们曾经的约定——“趁着还有机会，就该去探索更多不同的地方”，于是你选择留在相遇的地方开疆拓土，而我则选择来到了你成长的土壤。感谢你和你的家人，在我赴华求学期间给予的细心照料与温暖。

最后，浙大的求学之旅即将落下帷幕，但“求是创新”的校训精神将永远伴随着我。即便学生生涯即将告一段落，我依然会将学习作为我毕生的渴望与追求。凡事求真求变，永不设限，愿我们都能做个有趣的人。

摘 要

在直播电商从“流量驱动”向“内容驱动”转型的背景下，主播如何通过语言策略有效提升转化率成为关注焦点。现有研究多关注指令性话术（如催单、限时）的即时刺激作用，却往往难以解释直播销售中常见的“时间错位”与“爆发式增长”现象。本研究基于叙事传输理论（Transportation Theory）与心理模拟机制，提出“叙事型销售（Storytelling）”这一新视角，旨在揭示主播叙事行为对消费者购买决策的动态影响机制。本研究以淘宝女装直播间的分钟级高频数据为样本，通过对主播话术进行逐分钟编码，识别出包含场景构建、体验描述及问题解决链条的“叙事型片段”，并结合弹幕情感分析与销售数据，运用最小二乘回归（OLS）、向量自回归（VAR）及动态时间规整（DTW）等方法进行实证检验。研究结果表明：（1）叙事型话术不具有显著的“即时销售效应”，这意味着故事的听取与消化需要时间，其对销售的影响是非同步的；（2）叙事显著降低了决策摩擦，通过稳定观众情绪（降低负向弹幕比例），显著预测了随后数分钟内的销售增长，呈现出明显的“滞后效应”；（3）叙事是“爆发式销售”的催化剂，相比于平滑的销量增长，高强度的产品相关叙事更能预测随后的销售脉冲（Sales Spikes）。本研究通过引入时间维度，证实了叙事在直播中并非简单的信息传递，而是一种“时间结构化”的说服机制：它通过前期的内容铺垫与心理模拟，在叙事完成后集中释放购买力。这一结论修正了直播营销中传统的“即时刺激-反应”偏见，为“内容电商”的长期价值提供了微观层面的实证证据。

关键词：直播电商；叙事型话术；心理模拟；时间动态；滞后效应

Abstract

As livestream commerce evolves from a traffic-driven to a content-driven model, understanding how streamer linguistic strategies drive conversion has become critical. While existing literature often focuses on the immediate stimulus of directive language (e.g., Call-to-Action), it struggles to explain the "temporal misalignment" and "burst-like" sales patterns frequently observed in high-frequency data. Drawing on Transportation Theory and the mechanism of mental simulation, this study adopts a storytelling-based perspective to investigate the dynamic impact of narrative strategies on consumer purchasing behavior. Using minute-level data from Taobao women's apparel livestream sessions, this study quantifies "storytelling" by identifying linguistic segments containing scenario construction, experiential descriptions, and problem-solution chains. By integrating these measures with real-time sentiment analysis and sales data, the study employs Ordinary Least Squares (OLS), Vector Autoregression (VAR), and Dynamic Time Warping (DTW) to examine temporal dynamics. The empirical results indicate that: (1) Storytelling does not generate a significant contemporaneous effect on sales, suggesting that narrative processing requires time and its influence is non-synchronous; (2) Storytelling functions as a mechanism for reducing decision friction; by stabilizing audience sentiment (reducing negative feedback), it significantly predicts sales growth in subsequent minutes (lagged effect); and (3) Storytelling acts as a catalyst for "sales bursts," predicting sudden spikes in demand significantly better than it predicts average sales. By incorporating a temporal dimension, this study demonstrates that storytelling is not merely information transmission but a "temporal structuring" mechanism of persuasion: it builds purchase reasoning through mental simulation, leading to concentrated conversion after the narrative concludes. These findings challenge the traditional "stimulus-response" assumption in livestream marketing and provide micro-level empirical evidence for the value of content-based selling.

Keywords: Livestream Commerce; Storytelling; Mental Simulation; Temporal Dynamics; Lagged Effect

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1 INTRODUCTION

1.1 Research Background

Livestream commerce has fundamentally reshaped the digital retail landscape by integrating real-time interaction, product demonstration, and transaction execution into a synchronized virtual environment. In markets such as China, platforms like Taobao Live have evolved from experimental marketing channels into dominant drivers of gross merchandise value (GMV), particularly in non-standardized categories such as apparel. Unlike traditional e-commerce, which relies on static text and asynchronous browsing, livestreaming enables streamers to communicate directly with audiences, creating a high-frequency feedback loop of linguistic cues and emotional reactions.

However, as the industry matures, the effectiveness of traditional “hard-sell” strategies, characterized by repetitive Call-to-Action (CTA) prompts, high-pressure scarcity signals, and aggressive volume has begun to diminish. Empirical observations suggest that while such directive language aims to trigger immediate purchases, it often induces psychological reactance among viewers, leading to engagement fatigue rather than conversion. Consequently, the industry is witnessing a fundamental shift from "traffic-driven" promotion to “content-driven” engagement. In this new context, storytelling has emerged as a critical linguistic strategy. Rather than merely listing product attributes, streamers increasingly rely on narrative techniques such as constructing usage scenarios, sharing personal experiences, and articulating problem-solution chains to facilitate consumer understanding and reduce the uncertainty inherent in online shopping.

Despite this practical shift, the theoretical understanding of how storytelling influences sales performance in real-time remains limited. High-frequency data indicates that sales in livestreaming do not follow a linear “stimulus-response” pattern but instead exhibit temporal misalignment and burst-like characteristics. This suggests that the persuasive effect of storytelling is not instantaneous but operates through a dynamic process of cognitive processing and emotional accumulation, a phenomenon that requires rigorous empirical investigation.

1.2 Problem Statement

While livestream commerce has attracted significant scholarly attention, existing research has predominantly focused on static determinants of sales, such as aggregate trust, social presence, or streamer attractiveness. Studies that do examine linguistic strategies often treat streamer speech as a homogenous variable or focus exclusively on directive promotional cues (e.g., CTA frequency). This approach presents three critical limitations that this thesis seeks to address.

First, there is a lack of granular investigation into narrative-based persuasion. Current literature has not sufficiently quantified "storytelling" as a distinct linguistic construct in livestreaming. Specifically, it remains unclear how narrative intensity defined by the presence of characters, plots, and experiential details differs from standard transactional speech in its impact on consumer decision-making.

Second, the mechanism of uncertainty reduction is underexplored. For experience goods like women's apparel, consumers face high decision friction due to the inability to physically inspect products. While theory suggests that storytelling facilitates mental simulation (i.e., imagining product usage), empirical evidence linking this cognitive mechanism to actual sales outcomes in a live environment is scarce.

Third, methodological approaches often fail to capture temporal dynamics. Most studies employ contemporaneous regression models that assume linguistic cues produce immediate effects. However, narrative persuasion is inherently temporal; it requires time to unfold and time for the audience to process. Preliminary evidence suggests that sales peaks often occur after a narrative episode concludes, resulting in a lagged effect that static models cannot detect. This study argues that to fully understand livestream conversion, one must adopt a dynamic perspective that accounts for the time lags and non-linear relationships between streamer storytelling, audience sentiment, and sales bursts.

1.3 Research Questions

This study investigates how the linguistic choices of streamers specifically their use of narrative techniques that affect sales outcomes over time. To address the

theoretical and practical gaps identified earlier, the current research seeks to answer four specific questions:

- RQ1: How can we effectively measure storytelling within the fast-paced environment of livestreaming, and how does its usage pattern differ from traditional promotional calls-to-action?
- RQ2: Does a streamer's narrative intensity act as an environmental stabilizer, specifically by reducing the volatility of audience sentiment in the live chat?
- RQ3: Are the financial returns of storytelling immediate or delayed? In other words, does narrative intensity better predict future sales rather than current-minute sales?
- RQ4: Can storytelling account for the sudden, concentrated sales spikes often seen in these broadcasts, and does it structurally align with consumer purchasing behavior over a timeline?

1.4 Research Objectives

Corresponding to the questions above, the primary goals of this thesis are as follows:

First, the study aims to establish a reliable coding framework to quantify narrative speech from raw minute-level transcripts, separating elements like scenario building from simple promotional noise. Second, it tests the delayed persuasion effect using time-series econometrics. By applying Distributed Lag Models and Vector Autoregression, the research examines if storytelling impacts the subsequent ($t + k$) window rather than generating an immediate reaction. Third, the project explores the stabilizing function of narratives on audience emotion, assessing whether stories lower decision friction without relying on traditional, static mediation assumptions. Finally, the study evaluates whether narrative density can predict extreme sales bursts, thereby confirming its role in structuring temporal conversion.

1.5 Scope of the Study

The empirical context for this investigation is the women's apparel sector on Taobao Live. Apparel represents a classic experience good where physical touch is

impossible online, making verbal description highly relevant to the consumer's decision-making process.

The analysis relies on a high-frequency, minute-by-minute dataset collected from a top-tier merchant. While the raw data contained 5,836 observations, the cleaning process yielded a final working sample of 5,000 valid minutes. This dataset merges three distinct streams of information: the streamer's spoken text, the real-time viewer comments (danmaku) used to quantify sentiment, and the backend transaction records.

It is important to note that this study only looks at verbal and textual exchanges. Non-verbal factors, such as the streamer's appearance, vocal tone, or visual set design, are excluded. The focus is strictly on within-session, micro-temporal dynamics rather than long-term brand equity building across multiple broadcasts.

1.6 Structure of the Thesis

The thesis is organized into six chapters:

- Chapter 1 provides the background and outlines the core problem.
- Chapter 2 reviews the literature on livestream marketing, language strategies, and the psychology of narrative transportation.
- Chapter 3 constructs the theoretical framework, detailing the "Simulation-Stabilization-Conversion" pathway and proposing hypotheses H1 through H4.
- Chapter 4 documents the data collection procedures and the specific econometric models (OLS, VAR, and DTW).
- Chapter 5 reports the empirical results, beginning with descriptive patterns and moving into the time-series estimations.
- Chapter 6 discusses the theoretical and practical implications of these findings.
- Finally, Chapter 7 concludes the thesis, notes its limitations, and suggests concrete avenues for future research.

2 LITERATURE REVIEW AND THEORETICAL FOUNDATIONS

2.1 Literature Review

2.1.1 The Evolution of Livestream Commerce: From Traffic to Content

2.1.1.1 From Static E-commerce to Real-time Interaction

Livestream commerce has fundamentally transformed digital retail by merging product demonstrations with real-time social interaction. Traditional e-commerce platforms primarily rely on asynchronous browsing, where consumers evaluate products through static images, text descriptions, and historical reviews. This static model often limits the richness of product information and isolates the consumer in the decision-making process. Livestreaming disrupts this model by creating a synchronous, highly interactive shopping environment. Scholars note that this format introduces a high-frequency loop of communication, allowing streamers and viewers to exchange linguistic cues, answer inquiries, and share emotional reactions in real-time (Wongkitrungrueng & Assarut, 2020; Sun et al., 2019). However, the rapid scaling of such real-time interactions can lead to severe congestion effects. Recent evidence from the Information Systems field (Zhao et al., 2025) demonstrates that as livestream viewer groups expand, the resulting topic incoherence and emotional volatility significantly diminish individual engagement. This highlights the critical necessity for streamers to employ structured narrative strategies to maintain audience focus and mitigate cognitive chaos.

In markets such as China, platforms like Taobao Live have moved beyond their experimental phases to become primary revenue drivers for retailers. As the format matures, recent empirical studies observe a structural shift in its core competitive advantage. While the initial phase of livestreaming heavily relied on “price incentives” and aggressive discounting to attract traffic, scholars highlight that sustainable profitability now fundamentally depends on immersive engagement and narrative content quality (Ma et al., 2025). This industry shift prompts a deeper academic inquiry into how streamer-generated content, interacting with platform affordances, actively shapes customer value co-creation rather than relying on mere price manipulation (Ning et al., 2025).

2.1.1.2 The Limits of "Hard-Sell" Tactics

Despite the industry's evolution toward content, a significant portion of livestreaming practice still relies on "Hard-Sell" tactics. This tactic is characterized by repetitive Call-to-Action (CTA) phrases (e.g., "Buy it now!"), high-pressure scarcity signals ("Only 5 left!"), and an aggressive vocal delivery intended to force impulse purchases (Chen, Yao, & Xu, 2021).

While hard-sell tactics can generate short-term excitement, recent literature suggests their effectiveness is rapidly diminishing. Gao et al. (2023) observe that an overabundance of promotional cues in a live broadcast often leads to "cognitive overload," accelerating viewer fatigue rather than sustained engagement. This observation is firmly grounded in Psychological Reactance Theory (Brehm, 1966), which posits that when consumers perceive an overt threat to their behavioral freedom such as being aggressively pushed to make a transaction. They experience psychological resistance and actively reject the persuasion attempt. Empirical evidence from influencer marketing supports this, showing that overtly aggressive selling compromises the broadcaster's perceived authenticity and significantly increases viewer churn rates (Audrezet et al., 2020). Consequently, both industry practice and academic discourse are shifting focus toward "content-driven" engagement. Recognizing that the qualitative depth of the presentation often outweighs the volume of promotional shouts, recent research identifies that building psychological ownership through engaging narratives is far more effective for maintaining long-term viewer loyalty than aggressive sales pressure (Shen et al., 2025; Zhou, Tong, & Su, 2022).

2.1.2 Linguistic Strategies and Consumer Behavior in Digital Marketing

2.1.2.1 The Role of Language Style in E-commerce

The impact of linguistic cues on consumer decision-making has been widely documented, but its application has recently shifted from text-based reviews to real-time synchronous video environments. While foundational studies established that affective and figurative language enhances product evaluations for experience goods (Kronrod & Danziger, 2013; Ludwig et al., 2013), recent literature specifically emphasizes the critical

role of streamer language styles in livestream commerce. Modern studies indicate that dynamic host communication strategies and language variations are primary drivers of consumer engagement and trust (Mubarok et al., 2024). Furthermore, Ma et al. (2025) note that the persuasive influence of a live streamer heavily depends on sophisticated language strategies rather than merely counting visual attributes or speaking speed. Despite these advancements, existing methodologies often treat streamer speech as a homogenous or aggregated variable, failing to capture the micro-level structural differences between narrative storytelling and directive promotional shouts during a continuous broadcast.

This shift in research focus is corroborated by the latest evidence in the Information Systems (IS) field. For example, He et al. (2025) utilized randomized field experiments to demonstrate that real-time adjustments in a streamer's linguistic tactics and improvisational pacing directly drive sales performance. Furthermore, the effectiveness of these linguistic choices is heavily contingent on the structure of in-consumption information cues, which have been found to significantly prolong viewer immersion and engagement (Sim et al., 2024). These contemporary studies suggest that the temporal distribution of narrative information is as critical as the content itself in the livestreaming context.

2.1.2.2 Directive vs. Narrative Persuasion

To address this theoretical gap, it is crucial to distinguish between directive and narrative persuasion strategies. Directive persuasion corresponds to the aforementioned "hard-sell" tactics, relying on explicit instructions and logical arguments designed to prompt immediate action. In contrast, narrative persuasion utilizes storytelling structures to convey information implicitly. Escalas (2007) defines narrative processing as a distinct cognitive state wherein consumers construct a mental story involving the product, rather than analytically evaluating isolated product attributes. While narrative persuasion is a well-established mechanism in traditional advertising (Elder & Krishna, 2012), its structural measurement and temporal effects within the synchronous, high-pressure environment of livestream commerce require specific empirical investigation.

2.1.2.3 Sensory Marketing and Mental Imagery Vividness

Building on the distinction between directive and narrative persuasion, contemporary literature increasingly emphasizes the role of sensory marketing and embodied cognition in digital retail environments. Mental imagery defined as a consumer's subjective experience of perceptually picturing products or events that are not physically present has emerged as a core mechanism for engaging consumers (van Laer et al., 2014). In the context of livestreaming, streamers act as substitutes for the consumers' physical senses. When a streamer utilizes descriptive language to simulate how a garment drapes or feels, they actively enhance imagery vividness. Recent empirical studies in live commerce (Li & Park, 2025) confirm that specific broadcast characteristics and vivid streamer descriptions directly stimulate mental simulation and vicarious experience, bridging the tactile gap inherent in online shopping. Furthermore, providing sensory-emotional appeals through immersive storytelling triggers dual pathways of flow and cognitive engagement, which recent evidence shows is crucial for long-term customer value co-creation (Chou & Chen, 2022). This conceptualization strongly supports the premise that storytelling is not merely a tool for information delivery, but a catalyst for deep mental simulation that requires cognitive internalization.

2.1.3 The Temporal Dynamics of E-Commerce Persuasion

A critical limitation in early livestreaming research was the assumption of a static, contemporaneous relationship between streamer inputs and consumer responses. However, contemporary literature increasingly acknowledges the temporal complexities of online consumer behavior. While foundational psychological theories (Lang, 2000) established that cognitive reflection unfolds over time, recent live-streaming research specifically highlights that interactive cues and narrative persuasion operate through a delayed temporal mechanism rather than instant stimulus-response (Fan et al., 2024). The concept of "processing latency" is critical, suggesting that consumers require a temporal buffer to mentally simulate narrative claims, mitigate perceived risks, and transition from an inspired state to a transactional action (Ning et al., 2025). Despite this theoretical consensus, methodological approaches capable of capturing these micro-temporal lags

remain scarce. Most existing research still relies on post-event surveys, bibliometric aggregates, or session-level data (Huang et al., 2024; Kang et al., 2021), which mathematically forces a simultaneous relationship between variables. By failing to account for high-frequency, minute-by-minute variance, the existing literature leaves the precise temporal structuring of persuasion and the phenomenon of delayed "sales bursts" largely unquantified.

2.1.4 Summary of Literature and Research Gaps

A review of the existing literature reveals significant theoretical advancements in our understanding of livestream commerce and narrative persuasion. However, several critical research gaps remain unaddressed:

First, there is a methodological reliance on static, aggregate analysis. The majority of livestreaming research utilizes session-level data (e.g., total GMV per broadcast) or cross-sectional surveys to measure persuasion outcomes. This approach inherently assumes a contemporaneous "stimulus-response" model. It fails to capture the micro-temporal dynamics of persuasion, specifically how a linguistic input at minute t influences behavioral outputs at minute $t + k$. Consequently, the "burst-like" sales patterns and delayed conversions frequently observed in industry practice remain largely unexplained in the academic literature.

Second, existing literature lacks a rigorous measurement of narrative intensity in real-time contexts. While the psychological mechanisms of storytelling are well-documented in traditional advertising (Escalas, 2007), empirical studies in livestreaming often reduce streamer language to basic keyword counts or crude sentiment scores. The literature lacks a granular framework capable of measuring the specific dimensions of mental simulation such as scenario construction and experiential elaboration within high-frequency Chinese e-commerce corpora.

Finally, the dynamic interaction between narrative pacing and emotional stabilization remains underexplored. Although emotional contagion is a recognized phenomenon in livestreaming, it is typically treated as an independent environmental variable. The temporal mechanism by which streamer storytelling actively suppresses decision friction to pave the way for sales spikes requires rigorous time-series verification.

This study aims to bridge these gaps. By utilizing minute-level transcripts, real-time sentiment analysis, and dynamic econometric modeling, this research transitions the analysis of livestream persuasion from a static framework to a temporally structured "simulation-stabilization-conversion" model.

2.2 Research Roadmap

To systematically address the research objectives, this study follows a rigorous empirical trajectory. The research workflow begins with theoretical deduction, followed by minute-level data crawling and text mining, leading into econometric estimations using DLM and VAR, and concluding with robustness checks.

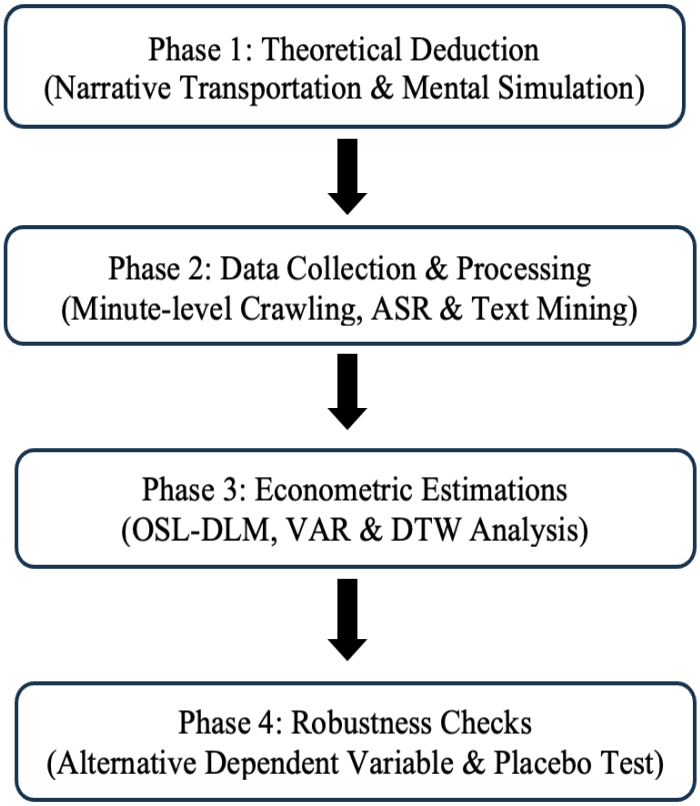


Figure 1.1 Research Roadmap of the Study

2.3 Theories of Narrative Persuasion and Cognitive Processing

2.3.1 Elaboration Likelihood Model vs. Narrative Transportation

To understand why narrative content facilitates sales particularly for experience goods, it is necessary to examine how consumers process incoming information.

Traditional marketing communication often relies on the "Central Route" of the Elaboration Likelihood Model (ELM) (Petty & Cacioppo, 1986). This route assumes a rational consumer who analytically evaluates facts, prices, and fabric specifications. While effective for standardized, utilitarian products, this logical approach often triggers "counter-arguing" when applied to non-standardized apparel. When a streamer lists objective facts, viewers naturally deploy cognitive defenses and scrutinize the claims (Dimoka et al., 2012).

Narrative storytelling overcomes this analytical resistance by triggering Narrative Transportation (Green & Brock, 2000). When a viewer becomes absorbed in a well-constructed story or scenario, their cognitive resources are shifted from critically evaluating the "sales pitch" to imagining the events unfolding within the narrative. A comprehensive meta-analysis by van Laer et al. (2014) confirms that this state of immersion effectively "suspends disbelief." By occupying the viewer's working memory with narrative simulation, the streamer minimizes the audience's capacity for counter-arguing, thereby disarming their psychological defenses against the underlying commercial intent (Moyer-Gusé, 2008). Recent empirical evidence strictly aligns with this classic view. For instance, Wang et al. (2026) demonstrate that in contemporary fast-paced live commerce, host storytelling significantly outperforms traditional static product displays by fostering deep narrative immersion, which directly weakens viewers' rational evaluation and facilitates impulsive purchasing.

2.3.2 Mental Simulation and Uncertainty Reduction

Beyond merely lowering psychological resistance, storytelling serves as a functional mechanism for uncertainty reduction. The primary barrier in online apparel shopping is the absence of physical inspection and tactile feedback (Spears & Yazdanparast,

2013). This sensory deficit creates a significant gap between digital presentation and physical reality, generating high decision friction.

Storytelling bridges this gap by facilitating Mental Simulation. When a streamer articulates a specific scenario such as “wearing this dress to a beach party” or describes a fabric as “feeling like a cloud”; they prompt the viewer to “vicariously try on” the product (Elder & Krishna, 2012). Feiereisen et al. (2008) describes this process as an “analogical transfer,” enabling consumers to mentally integrate the product into their own lived experiences. By translating abstract product features into concrete sensory and situational simulations, storytelling effectively lowers the perceived risk and friction associated with the purchase decision (Guo et al., 2021). Extending this mechanism to the modern digital context, Li and Park (2025) recently confirmed that streamer-guided mental simulation effectively generates a “vicarious experience” for consumers, thereby sharply reducing perceived online shopping risks and elevating actual purchase intentions.

2.3.3 The Temporal Nature of Cognitive Processing

A critical, yet frequently overlooked, dimension of mental simulation is its temporal requirement. Extracting meaning from a narrative and projecting oneself into a simulated scenario is a cognitively demanding process. According to the Limited Capacity Model of Motivated Mediated Message Processing (LC4MP) (Lang, 2000), human cognitive bandwidth is finite. When viewers allocate their cognitive resources to encode and process an immersive narrative, their capacity to simultaneously execute secondary tasks such as evaluating pricing or navigating a checkout interface is heavily constrained.

The cognitive burden inherent in this process is particularly obvious in high-density information environments. As Wang et al. (2025) identified, the simultaneous processing of video, audio, and interactive cues in livestreaming imposes substantial cognitive constraints on consumers. In such settings, narrative structures function as essential cognitive anchors; by providing a coherent framework, they help manage the viewer’s limited capacity while temporarily prioritizing mental simulation over immediate transactional actions. Consequently, processing complex narrative information under these high cognitive constraints requires a distinct temporal buffer.

This theoretical perspective suggests that the persuasive effect of storytelling cannot be instantaneous. The cognitive decoupling required for narrative processing implies that the actual behavioral response (purchasing) will structurally lag behind the linguistic stimulus, a dynamic that static e-commerce models fail to capture.

2.4 Social Presence and Emotional Contagion in Live Chats

2.4.1 Collective Emotion and Decision Friction

Livestreaming is uniquely characterized by its high Social Presence, primarily facilitated through the "Danmaku" (bullet screen) interface, which allows viewers to observe the reactions of others in real-time. While this feature fosters community building, it also creates an environment highly susceptible to Emotional Contagion (Hatfield, Cacioppo, & Rapson, 1994). In the context of experience goods, when product details are ambiguous, the live chat can rapidly accumulate negative or anxious inquiries (e.g., sizing doubts, quality concerns). Huang et al. (2023) conceptualize this as "collective anxiety." Furthermore, latest investigations into Danmaku interfaces reveal that the high-density scrolling comments amplify specific emotional expressions, making viewers highly susceptible to the collective emotional field co-created by the streamer and the audience (Recent studies in 2025 and 2026, e.g., MDPI, E-Commerce Letters). If unmanaged, this volatile emotional field acts as a severe form of social decision friction, paralyzing the purchasing intent of the broader audience.

2.4.2 The Dynamic Effect of Real-Time Interaction and Trust

While emotional contagion explains the rapid spread of sentiment, recent studies have begun to explore the dynamic effects of real-time interactivity on trust and decision friction. Cai et al. (2018) initially highlighted that livestream commerce fosters a "transcendent customer experience." Building on this, recent game-theoretic analyses by Fan et al. (2024) demonstrate that high-quality, real-time interactivity fundamentally transforms the live-streaming sales mode. In this ecosystem, the streamer's communication style directly dictates whether the audience forms cognitive trust (based on expertise) or affective trust. As Shen et al. (2024) recently confirmed, affective trust is rapidly

accelerated through parasocial relationships cultivated by the streamer's relational communication.

Additionally, excessive reliance on exaggerated or overly aggressive promotional language often found in hard-sell tactics has been shown to generate cognitive resistance. This dynamic was empirically validated by Ma et al. (2025), whose analysis of streamer language strategies revealed an inverted U-shaped relationship between exaggerated promotional intensity and product sales. Their findings indicate that extreme sales pressure eventually triggers consumer reactance and subsequent purchase regret, a concept rooted in classic reactance theory (Brehm, 1966). This underscores the necessity of a stabilizing communication strategy. Storytelling manages the emotional field by fostering affective trust rather than over-stimulating the audience, thereby systematically reducing the volatility of the chat environment.

2.4.3 Storytelling as an Environmental Stabilizer

Extant literature indicates that streamers must manage not only product information but also the psychological environment of the broadcast. We argue that storytelling functions as a critical environmental stabilizer. A well-crafted narrative aligns the scattered attention of the audience, providing a coherent "affective context" (Hu & Chaudhry, 2020). This regulatory mechanism is strongly supported by Mubarok et al. (2024), who found that adaptive host communication styles and language variations are essential for managing the volatile ecosystem of live-streaming platforms. Rather than allowing the chat room to be dominated by fragmented, individual doubts, a relatable story such as a streamer sharing a personal struggle with finding the right fit fosters empathy and cognitive resonance (Gong et al., 2020). Furthermore, Li and Park (2025) recently revealed that when streamers employ distinct broadcast characteristics to guide mental simulation, it creates a synchronized "vicarious experience" for the audience. This shared attentional synchronization limits the cognitive space available for viewers to process external distractions or negative noise. Consequently, storytelling serves a dual purpose: it persuades the individual through mental simulation and simultaneously stabilizes the collective emotional climate by mitigating sentiment volatility, thereby establishing the psychological safety necessary for conversion.

3 HYPOTHESIS DEVELOPMENT

3.1 Theoretical Framework Overview

Building upon the limitations of the "stimulus-response" model identified in the literature review, this chapter constructs a dynamic theoretical framework to explain the mechanism of livestream persuasion. While traditional e-commerce models often treat conversion as an instantaneous decision triggered by price or scarcity, we posit that narrative persuasion is a multi-stage temporal process.

To capture this complexity, we synthesize Narrative Transportation Theory (Green & Brock, 2000) and Mental Simulation Theory (Elder & Krishna, 2012) into a unified "Simulation-Stabilization-Conversion" Framework. This framework argues that the impact of streamer storytelling on sales is not direct but is mediated by two parallel psychological processes: cognitive simulation (which requires time) and emotional stabilization (which reduces friction).

3.1.1 The Logic of "Cognitive Trade-off" and Temporal Lag

The central premise of our framework is that storytelling imposes a "Cognitive Trade-off" on the viewer. Unlike a simple discount signal (e.g., "50% off"), which requires minimal cognitive effort to grasp, g a narrative requires the audience to engage in Mental Simulation.

When a streamer describes a usage scenario, the viewer must allocate cognitive resources to visualize this scene and project themselves into it. According to the Resource Allocation Theory, human attention is finite. Recent empirical findings strongly corroborate this logic in digital environments; for instance, Wang et al. (2026) confirmed that deep narrative immersion in live-streaming significantly occupies working memory, thereby temporarily delaying immediate impulse buying. During the immersion phase of the story, the viewer's cognitive capacity is consumed by the mental simulation, leading to a natural inhibition of the motor action required to navigate the checkout interface. This theoretical logic explains why we anticipate a structural lag between the linguistic input

and the behavioral output. The actual purchase decision can only be executed after the narrative concludes and cognitive bandwidth is released.

3.1.2 The Mechanism of "Emotional Anchoring"

Parallel to the cognitive process, our framework incorporates an environmental dimension. Livestreaming is characterized by high uncertainty and rapid emotional contagion (e.g., negative comments spreading anxiety about product quality). We integrate Social Presence Theory to propose that storytelling functions as an "Emotional Anchor."

By constructing a coherent and relatable context, the narrative aligns the collective attention of the audience, naturally suppressing the fragmented noise within the chat room. This dynamic is critical, as unmanaged "Danmaku" comments can rapidly escalate collective emotional anxiety and freeze consumer actions (You, 2026). In this context, specific host language strategies and communication styles serve as essential regulatory mechanisms to manage platform volatility (Mubarok et al, 2024). We argue that narrative stabilization does not magically create desire out of thin air; rather, it removes the social decision friction that prevents conversion. In our model, this reduction in sentiment volatility acts as a necessary precondition: a psychological clearing of the path that establishes the trust required for the subsequent release of purchasing power.

3.1.3 The "Accumulation-Release" Dynamic

Finally, combining these cognitive and emotional mechanisms, the framework predicts a non-linear outcome. Narrative persuasion functions similarly to charging a capacitor: purchasing intent is accumulated during the storytelling phase (while sales remain flat or suppressed) and is released collectively once the story ends. This theoretical deduction leads us to reject the assumption of linear sales growth in favor of a "Burst-Like" conversion model.

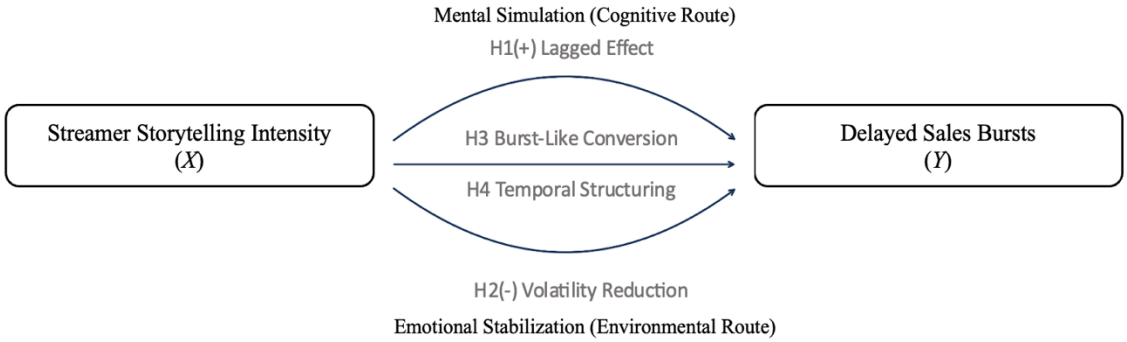


Figure 3.1 Theoretical Model and Hypotheses Framework

Note: The pathways denote the structural hypotheses derived from the theoretical framework, illustrating the dual routes leading to delayed sales bursts.

3.2 Hypotheses Development

Based on the theories of Narrative Transportation and Mental Simulation, and Temporal Structuring, this study proposes four hypotheses how storytelling drives sales over time.

3.2.1 Storytelling and Temporal Lag (H1)

Standard economic models often assume a contemporaneous stimulus-response relationship where marketing inputs lead to immediate sales. However, narrative persuasion deviates fundamentally from this linear pattern. To explain the temporal lag between streamer storytelling and viewer purchase, we draw upon the Limited Capacity Model of Motivated Mediated Message Processing (LC4MP) (Lang, 2000) alongside recent advancements in mental simulation (Li & Park, 2025). Human cognitive resources are finite. When a streamer engages in intensive storytelling, such as constructing a detailed usage scenario or describing the tactile sensation of a fabric, viewers must allocate significant cognitive resources to visualize and process this narrative. Recent literature on livestream interactivity emphasizes that high-quality informational cues inherently demand a longer cognitive internalization phase before manifesting as concrete engagement (Ning et al., 2025). During this active immersion phase, the cognitive resources required to evaluate pricing, navigate the interface, and execute a transaction are temporarily diverted. We conceptualize this as a state of “Processing Latency.” The viewer’s mind is occupied with experiencing the product virtually, leading to a temporary decoupling from the

physical act of buying. This theoretical deduction aligns with objective evidence from recent field data. Empirical investigations have confirmed that specific linguistic delivery and the pacing of information cues can fundamentally intervene in the consumer's decision-making cycle (He et al., 2025; Sim et al., 2024). These studies demonstrate that high-engagement cues can lengthen the time spent in the consumption phase before a purchase occurs. Consequently, we argue that the cognitive immersion induced by storytelling creates a necessary temporal buffer, where the purchase intention is accumulated rather than immediately discharged.

- ***H1: Streamer storytelling intensity has a positive lagged effect on sales conversion.*** The positive impact of narrative intensity on conversion rates manifests in subsequent minutes ($t+k$) rather than contemporaneously (t).

3.2.2 Storytelling as an Emotional Stabilizer (H2)

Livestreaming environments are characterized by high social presence and rapid emotional contagion. For experience goods like apparel, high uncertainty often triggers a fragmented and volatile emotional field in the live chat. This high Sentiment Volatility signals severe decision friction, which deters collective purchasing behavior. Recent scholarly discourse highlights that real-time interactivity not only transmits information but fundamentally restructures the psychological environment, serving as a critical antecedent for building psychological ownership and consumer loyalty (Shen et al., 2024). We posit that storytelling functions as a dynamic environmental stabilizer. Rather than merely conveying product parameters, a well-structured narrative anchors the collective attention of the audience. By immersing viewers in a shared affective context, storytelling effectively suppresses the chaotic noise within the "Danmaku" (bullet screen) and mitigates the psychological reactance frequently induced by aggressive hard-sell tactics (Ma et al., 2025). From a dynamic systems perspective, this stabilization process is a critical precondition that actively clears the path for subsequent conversion.

- ***H2: Streamer storytelling intensity has a negative effect on audience sentiment volatility.*** High narrative intensity dynamically suppresses the fluctuation of audience sentiment, creating a stabilized emotional environment that facilitates subsequent sales.

3.2.3 Storytelling and Burst-Like Conversion (H3)

A distinct feature of livestreaming is that transactions rarely occur in a smooth, continuous flow. Instead, they happen in sudden, concentrated bursts. To explain why storytelling specifically drives these spikes, we must look at how narratives structure audience attention. When a streamer merely lists basic product features, viewers tune in and out randomly. However, telling a relatable story forces the audience to follow a specific chronological plot. Recent studies on livestream interactivity (Sim et al., 2024) reveal that the temporal structure of information delivery is vital for sustaining immersion. Because a compelling story aligns the psychological momentum of thousands of viewers simultaneously, their purchasing intent is temporarily held back while they listen. Shen et al. (2025) highlight that this shared affective state synchronizes viewer behavior. Therefore, the conversions do not occur gradually; they flood in the moment the story resolves and the psychological tension is released. This implies that high-intensity storytelling will not just lift average sales slightly, but it will act as the primary trigger for extreme sales spikes.

- ***H3: Streamer storytelling intensity positively drives the occurrence of burst-like sales.*** Storytelling explains the variance in extreme sales spikes significantly better than it explains fluctuations in average linear sales.

3.2.4 Temporal Structuring of Persuasion (H4)

The underlying logic of mental simulation requires us to rethink the timeline of online persuasion. In traditional static e-commerce, the “stimulus” (reading a product description) and “response” (clicking buy) can happen almost simultaneously. However, Wang et al. (2025) recently highlighted that the synchronous, high-density environment of live commerce places severe capacity constraints on consumers. Because human cognitive bandwidth is strictly limited, viewers cannot deeply process a narrative scenario and

navigate a complex checkout interface at the exact same second. The cognitive effort required to digest sensory descriptions, which Elder and Krishna (2012) term "embodied mental simulation", forces the consumer into a sequential processing mode. The rationale to purchase is actively built during the storytelling phase, but the physical act of buying must logically wait until the simulation ends. As a result, we expect a structural time lag in the broadcast data. The sales peaks will not perfectly mirror the narrative peaks in real-time; rather, they will systematically lag, clustering in the immediate post-narrative window.

- ***H4: Consumer purchasing behavior structurally lags behind streamer storytelling intensity.*** A structural time lag exists between the storytelling trajectory and the sales trajectory, where sales peaks systematically cluster in the post-narrative window.

Table 3.1 Summary of Hypotheses and Analytical Framework

Hypothesis	Theoretical Mechanism	Observed Consequence	Analytical Evidence
H1: Storytelling and Temporal Lag	**Narrative Transportation Theory:** Narrative processing requires cognitive resource allocation, creating a temporary "decoupling" from the immediate buying action.	Sales effect is ***delayed*** and non-synchronous, not immediate.	Significant positive coefficient for ***Lagged Sales*** ($Sales_{t+k}$) vs. insignificant coefficient for ***Contemporaneous Sales***($Sales_t$).
H2: Emotional Stabilization	**Uncertainty Reduction & Social Presence:** Storytelling constructs a relatable context that mitigates Decision Friction and stabilizes the emotional contagion in the chat.	Storytelling leads to a ***reduction*** in sentiment volatility or negative comments.	Mediation Analysis: Storytelling → ↓ Negativity → ↑ Sales.

Table 3.1 (continued) Summary of Hypotheses and Analytical Framework

H3: Burst-Like Conversion	**Synchronization Mechanism:** Unlike linear information, narratives align audience attention, leading to a collective release of stored purchasing power upon the story's climax.	Sales occur in concentrated ***bursts*** (spikes) rather than linear growth.	Storytelling intensity predicts ***Sales Bursts*** significantly better than it explains ***Average Sales***.
H4: Temporal Structuring	**Mental Simulation:** The "Reason to Buy" is constructed during the simulation (story), but the "Act of Buying" is a post-simulation behavior.	The sales trajectory follows the narrative trajectory with a ***systematic delay***.	***DTW (Dynamic Time Warping)*** reveals a consistent, non-linear time lag between linguistic input and behavioral output.

DATA AND METHODOLOGY

4.1 Research Design Overview

To empirically validate the theoretical framework proposed in Chapter 3 specifically the transition from "Argumentative Persuasion" to "Narrative Simulation", this study adopts a quantitative research design utilizing high-frequency, minute-level time-series data.

Unlike traditional cross-sectional studies that treat a livestream session as a static unit of analysis, this research conceptualizes livestreaming as a continuous temporal flow. The core methodological challenge is to capture the micro-dynamics of persuasion: how the linguistic input (Streamer's Storytelling) at time t structurally influences the behavioral output (Sales and Engagement) at time $t+k$.

The empirical strategy proceeds in three sequential stages. First, we process the unstructured textual data by converting raw streamer transcripts and real-time *danmaku* logs into quantifiable variables specifically, Narrative Intensity and Sentiment Volatility through content analysis. Second, we construct a unified minute-level panel dataset by merging these linguistic metrics with backend transaction logs. Finally, we employ a dynamic modeling approach: utilizing Ordinary Least Squares (OLS) with distributed lags to test linear delays, Vector Autoregression (VAR) to account for the endogeneity between chat engagement and sales, and Dynamic Time Warping (DTW) to visualize non-linear structural lags.

4.2 Data Collection and Processing

4.2.1 Empirical Setting

The data for this study was collected from Taobao Live, the leading livestream commerce platform in China. We selected a top-tier merchant in the Women's Apparel category as the focal case.

The rationale for this selection is twofold. First, apparel is a typical "experience good" characterized by high non-standardization. Consumers face significant uncertainty regarding fit and tactile feeling, making "Mental Simulation" via storytelling a critical

conversion mechanism. Second, this specific merchant employs professional streamers known for distinct narrative styles, providing sufficient variance in the independent variable.

4.2.2 Data Source and Cleaning

The observation period spans 14 consecutive days from March 15, 2024, to March 28, 2024. This specific timeframe was chosen to avoid major platform-wide promotional festivals (such as “618” or “Double 11”), thereby allowing us to isolate the specific effect of streamer linguistic strategies from external traffic shocks. Furthermore, the selected period represents a mature and stabilized phase of the Taobao Live ecosystem, where content-driven strategies have fully replaced early demographic dividends. This temporal selection ensures that the empirical patterns of narrative persuasion observed are highly representative of current market dynamics and are not confounded by the abnormal traffic spikes typically seen during the industry’s volatile introductory phase.

To clarify the sample structure, the raw data were compiled from a single, top-tier Taobao Live women’s apparel account hosting daily broadcast sessions (totaling 14 complete livestream sessions). Selecting a single merchant account over continuous sessions ensures that unobserved streamer-specific and brand-specific fixed effects are rigorously controlled for. The data were derived from two primary sources: backend transaction logs and real-time livestream archives. The initial transaction dataset comprised approximately 140,542 individual purchase records, capturing every transaction event. To capture the linguistic environment, we utilized Automatic Speech Recognition (ASR) technology to transcribe the 14 days of video content, encompassing both the streamer’s spoken script and the audience’s bullet-screen comments (danmaku).

To construct a unified time-series panel, we aggregated these distinct data streams into minute-level intervals. Specifically, transaction metrics such as total payment amount (*sum_div_pay_amt*), unique visitors (*count_visitor_uv*), and average purchase price were summed or averaged per minute based on their exact timestamps. Simultaneously, the textual data was segmented into corresponding minute blocks to strictly align with the sales data. For example, all words spoken between 20:00:00 and 20:00:59 were matched exclusively with the sales occurring within that identical interval.

Following the merger of the transaction and linguistic datasets, a rigorous cleaning procedure was implemented. We excluded observations containing technical interruptions (e.g., black screens), intervals with zero traffic typical of stream initiations and conclusions, and non-transactional segments such as streamer breaks. This process yielded a final, robust panel of 5,000 minute-level observations, providing a rigorous foundation for the subsequent dynamic time-series analysis.

Table 4.1 Sample of Processed Minute-level Data Format

Timestamp	*Story_t*	*Vol_t*	*Ln_Price*	*Ln_Like*	*Promo_t*	*Ln_CVR*
20:00:00	1025.000	0.751	0.000	0.000	1	-8.517
20:01:00	1089.000	0.812	0.000	0.693	1	-8.432
20:02:00	1156.000	0.654	0.231	0.693	1	-8.105
20:03:00	985.000	0.890	0.231	1.098	0	-8.921
20:04:00	920.000	0.912	0.154	0.000	1	-9.015
20:05:00	1050.000	0.720	0.154	0.431	1	-8.245
20:06:00	1205.000	0.610	0.510	1.386	0	-7.850
20:07:00	1180.000	0.635	0.510	1.609	1	-7.620
20:08:00	1010.000	0.780	0.000	0.693	1	-8.410
20:09:00	995.000	0.820	0.000	0.000	1	-8.550

Note: his table presents a 10-minute snapshot of the cleaned dataset. All variables follow the definitions and measurements detailed in Section 4.3.

4.3 Variable Measurement

4.3.1 Dependent Variable: Conversion Efficiency (*Ln_CVR*)

To rigorously measure the effectiveness of streamer persuasion while controlling for external traffic fluctuations, this study adopts Conversion Rate (CVR) as the primary dependent variable rather than Gross Merchandise Value (GMV). While GMV is a standard metric in e-commerce, it is heavily confounded by the volume of incoming traffic (Unique Visitors); a spike in GMV may simply reflect a platform-level traffic allocation rather than the streamer's ability to convert viewers. By contrast, CVR isolates the persuasion efficiency of the content, defined as the probability that a viewer converts into a buyer within a specific minute.

We calculated the minute-level CVR by dividing the Total Payment Amount (*sum_div_pay_amt*) by the number of Unique Visitors (*count_visitor_uv*) at minute *t*. Given that conversion rates in high-frequency livestreaming contexts typically follow a highly right-skewed distribution (with many minutes showing zero or low conversions), we applied a natural logarithm transformation to normalize the data structure for regression analysis. The final variable is measured as $Ln_CVR_t = Ln\ (CVR_t + \epsilon)$, where a small constant ϵ is added to handle zero-value observations, ensuring statistical robustness.

4.3.2 Independent Variable: Storytelling Intensity (*Story_t*)

To move beyond the limitations of simple keyword counting, this study adopts a semantic coding approach to measure "Storytelling". Unlike directive promotional speech (e.g., "Buy it now," "Last 30 seconds"), narrative speech is defined by its ability to transport the listener into a simulated reality. Drawing on Escalas's (2004) narrative structure framework, we classify streamer speech into a "Storytelling" variable only when it contains explicit elements of Scenario Construction, Experiential Description, or Problem-Solution Logic.

To apply this framework, the raw transcripts were segmented by minute. A linguistic segment was coded as a narrative unit if it successfully constructed a "mental simulation" for the viewer. To ensure the construct validity of this measure, we established a rigorous coding protocol. Table 4.1 provides concrete examples from our dataset, illustrating the distinction between high-intensity narrative segments and standard transactional speech.

Table 4.2 Exemplar Coding of Narrative Dimensions from Streamer Scripts

Narrative Dimension	Streamer Transcript Excerpt (Chinese/English Translation)	Coding Logic & Mechanism
Scenario Construction *(Contextual Simulation)*	**CN:** "姐妹们，想象一下，过年回家参加同学聚会，前男友就坐在对面，你穿这件大衣推门进去，那个气场瞬间就炸了..." **EN:** "Sisters, imagine this: You go home for the Chinese New Year reunion, and your ex-boyfriend is sitting right there. You walk in wearing this coat, and your aura instantly explodes..."	The streamer establishes a specific *time* (New Year), *character* (Ex-boyfriend), and *setting* (Reunion), triggering the viewer to simulate a social victory scenario.
Experiential Description *(Sensory Stimulation)*	**CN:** "这个面料摸起来不是那种死板的硬，它是像婴儿皮肤一样，软糯糯的，穿在身上就像云朵包着你，完全没有束缚感。" **EN:** "This fabric doesn't feel stiff or rigid; it feels like a baby's skin—soft and glutinous. Wearing it feels like being wrapped in a cloud, with absolutely no restriction."	Uses metaphorical sensory language ("baby skin," "cloud") to translate tactile information into a visual/cognitive experience, reducing uncertainty about fabric quality.
Problem-Solution *(Conflict & Resolution)*	**CN:** "很多梨形身材的宝宝不敢穿紧身裤，因为怕显胯宽。但这件裤子做了特殊的双褶设计，刚好把你最想藏的肉肉全都‘吃’进去了。" **EN:** "Many pear-shaped babies are afraid to wear skinny jeans because they fear showing wide hips (Conflict). But these pants use a special double-pleat design that specifically 'eats up' the flesh you want to hide most (Resolution)."	Structures the speech as a narrative arc: identifying a specific conflict (pain point) and offering the product as the narrative resolution.
Non-Narrative Speech *(Transactional / Directive)*	**CN:** "来，三二一，上链接！最后五十单，拍完就没有了，抓紧时间！ **EN:** "Okay, 3-2-1, link is up! Last 50 orders, once they're gone, they're gone. Hurry up!"	Purely directive. Contains no descriptive or simulation elements. Coded as Promotional Intensity (*Promo_t*), not Storytelling.

Note: Transcripts are extracted from the empirical dataset of a top-tier Taobao Live apparel merchant (March 2024). The coding process strictly distinguishes between “High-Narrative” segments (Categories 1-3) and “Low-Narrative” segments (Transactional/Control).

Based on the coding scheme illustrated above, Narrative Intensity (*Story_t*) is calculated as the cumulative volume of semantic units falling into the three narrative categories within minute *t*. Specifically, we utilize a Bag-of-Words (BoW) approach weighted by semantic relevance. Rather than a binary count (story vs. no story), *Story_t* represents the continuous density of narrative information:

$$Story_t = \sum (Freq_{Scenario} + Freq_{Experience} + Freq_{Solution}) \quad (1)$$

(Formula constructed based on the narrative framework of Escalas, 2004)

In this specification, Freq represents the frequency of validated narrative keywords and phrases (e.g., "imagine," "feels like," "party," "solution"). This continuous measure allows us to capture the true intensity of the storytelling effort, thereby enabling us to distinguish between a brief mention (low intensity) and an elaborate, immersive story (high intensity). It is worth noting that during preliminary data testing, we observed these three narrative dimensions (scenario construction, experiential description, and problem-solution) frequently co-occur in a streamer's continuous speech. Isolating them as separate independent variables for regression caused severe multicollinearity, which distorted the coefficient estimates. Consequently, utilizing a combined aggregate index more accurately reflects the streamer's overall narrative effort while ensuring model stability.

4.3.3 Intermediate Variable: Sentiment Volatility (*Vol_t*)

To test the hypothesis that storytelling facilitates conversion by stabilizing the viewing atmosphere (H2), we focused on the collective emotional stability of the audience rather than simple sentiment valence. In a livestreaming environment, the "Danmaku" (bullet screen) reflects the real-time cognitive state of the viewers. We argue that the dispersion of sentiment rather than its average positivity. It is a more accurate indicator of "decision friction."

We first processed the real-time Danmaku comments using SnowNLP, a Python-based natural language processing library specifically trained on Chinese e-commerce corpora. Each comment was assigned a sentiment score ranging from 0 (negative) to 1 (positive). Sentiment Volatility (*Vol_t*) was then measured as the standard

deviation of these sentiment scores within minute *t*. Theoretically, a high standard deviation indicates a fragmented emotional field characterized by conflicting opinions (e.g., a mix of praise, doubt, and inquiries), representing high noise and uncertainty. Conversely, a low standard deviation suggests a unified, consensus-driven environment. We posit that this reduction in volatility is a necessary precursor to collective purchase behavior.

4.3.4 Control Variables

To isolate the net effect of storytelling and mitigate potential omitted variable bias, this study incorporates several control variables that capture market fluctuations, promotional incentives, and baseline channel characteristics. First, we control for the average selling price (*Ln_Price*), calculated as the natural logarithm of the average transaction price per minute. Given the high price sensitivity inherent in the apparel category, price fluctuations can significantly dictate conversion rates independent of narrative persuasion.

Second, to account for real-time engagement and hard-sell tactics, we include a promotional shock indicator (*Promo_t*) and a social engagement metric (*Ln_Like*). *Promo_t* is a indicator variable set to 1 if explicit promotional keywords such as "Lottery" (抽奖) or "Seconds Kill" (秒杀) appear in the script, allowing us to distinguish between rational, price-driven purchases and emotional, narrative-driven conversions. Meanwhile, *Ln_Like* captures the log-transformed minute-level increment in likes. High social engagement frequently signals "social proof" or herd behavior, which can boost sales regardless of content quality. Finally, to control for the channel's accumulated trust capital and historical constraints, we include the streamer's follower base (*Ln_Fans*), measured by the log-transformed total fan count.

4.4 Analytical Strategy and Model Specification

To fully address the research questions and ensure robustness, this study employs a triangulated methodological approach combining linear and non-linear time-series techniques. We specifically employ the DLM to quantify the linear lag coefficients and the moderating effects of price (testing H1), while utilizing the VAR model to capture endogenous dynamic interactions and impulse responses (testing H2 and H3). Combining

these two approaches provides a more comprehensive view of the micro-dynamic processes than relying on a single VAR specification.

4.4.1 Ordinary Least Squares with Distributed Lags (OLS-DLM)

To verify H1 (Temporal Lag), a standard contemporaneous regression is insufficient because narrative processing requires a temporal investment from the viewer. We therefore employ a Distributed Lag Model (DLM). This specification allows the dependent variable to be affected by both current and past values of the independent variable.

Furthermore, to test the boundary conditions of narrative persuasion, we extend the model to include an interaction term between Storytelling and Price. We posit that storytelling serves as a "premium justification" mechanism, making it more effective for high-priced products.

The expanded model specification is:

$$Ln_CVR_t = \alpha + \sum_{k=0}^{n} \beta_k Story_{t-k} + \gamma(Story_{t-1} \times Price_t) + \delta Controls_t + \epsilon_t \quad (2)$$

(Empirical model specification adapted from livestreaming time-series analyses; e.g., He et al., 2025)

In this specification, β_k captures the direct effect of narrative intensity *k* minutes prior to the transaction, while the coefficient γ isolates the moderating effect of price. To empirically test the boundary conditions of narrative persuasion, our primary focus is on the term γ. A positive and statistically significant γ would indicate that storytelling serves as a premium justification mechanism, proving to be particularly effective for high-value conversions.

4.4.2 Vector Autoregression (VAR)

To test H2 and H3 while simultaneously accounting for the inherent endogeneity within livestreaming environments (e.g., the bidirectional relationship where high sales might also trigger high excitement), we employ a Vector Autoregression (VAR) framework. Unlike single-equation models that assume a strict one-way causality, the VAR approach treats the streamer's narrative input, the audience's emotional state, and the final

conversion outcome as an interconnected dynamic system, allowing us to capture complex feedback loops over time.

The VAR(p) model is specified as follows:

$$Y_t = C + A_1 Y_{t-1} + \cdots + A_p Y_{t-p} + E_t \quad (3)$$

(Standard Vector Autoregression specification applied to interactive settings; e.g., Fan et al., 2024)

In this specification, Y_t denotes the vector of endogenous variables, A_1 through A_p represent the matrices of autoregressive coefficients capturing the distributed lagged effects, and E_t designates the vector of error terms, or structural shocks, within the system. We construct the endogenous vector containing the three core dynamic variables, ordered based on our theoretical causal chain:

$$Y_t = [Story_t,\ Vol_t,\ Ln_CVR_t] \quad (4)$$

(Endogenous vector specification based on the theoretical framework in Chapter 3)

This Cholesky ordering fundamentally assumes that the streamer's narrative intensity ($Story_t$) acts as the exogenous input that instantaneously affects the system. This is followed by the intermediate variable, sentiment volatility (Vol_t), which subsequently influences the conversion rate (Ln_CVR_t) as the final outcome.

To trace the dynamic transmission of shocks within this system and validate our hypotheses, we utilize Impulse Response Functions (IRF). For the emotional stabilization mechanism (H2), we examine the response of sentiment volatility (Vol_t) to a one-standard-deviation orthogonalized shock in storytelling. Empirical support for H2 would manifest as a statistically significant negative response following the shock, confirming that an increase in narrative intensity effectively dampens the audience's fragmented emotions and reduces decision friction. Similarly, to validate the burst-like conversion hypothesis (H3), we trace the response of the conversion rate (Ln_CVR_t) to the same narrative shock. Rather than observing an immediate linear uplift typical of traditional promotional models, we anticipate a delayed, hump-shaped spike in the IRF

trajectory. This specific pattern would substantiate the premise that the persuasive impact of storytelling accumulates over time before releasing in a concentrated burst.

4.4.3 Dynamic Time Warping (DTW)

Finally, to validate H4 (Temporal Structuring) and quantify the "structural misalignment" between narrative input and sales output, we utilize Dynamic Time Warping (DTW). Standard correlation analysis relies on Euclidean matching (aligning time *i* with time *i*), which often fails to capture processes with variable speeds or delays.

DTW solves this by finding the optimal non-linear alignment path between the standardized Storytelling Sequence (*S*) and Sales Sequence (*Q*). The algorithm minimizes the cumulative distance:

$$DTW(S,Q) = \min \left\{ \sum_{k=1}^{K} w_k \right. \qquad (5)$$

(Standard Dynamic Time Warping algorithm specification

By analyzing the Warping Path, we can determine the synchronization pattern. If the optimal path systematically deviates from the diagonal towards the sales axis (indicating that the sales index *j* is consistently larger than the storytelling index *i*), it serves as mathematical evidence that the "Act of Buying" structurally follows the "Reason to Buy."

5 DATA ANALYSIS AND EMPIRICAL RESULTS

This chapter presents the empirical findings of the study, structured to systematically validate the proposed theoretical framework regarding the dynamic impact of streamer storytelling on sales conversion. The analysis begins with a detailed descriptive examination of the high-frequency dataset, followed by a rigorous correlation analysis to ensure model stability. Subsequently, the study employs a triangulated methodological approach: combining Ordinary Least Squares (OLS) with interaction terms, Vector Autoregression (VAR), and Dynamic Time Warping (DTW) to test the hypotheses regarding temporal lags, emotional stabilization, and structural alignment.

5.1 Descriptive Analysis

5.1.1 Sample Overview

Building upon the data collection and rigorous cleaning procedures detailed in Chapter 4, this empirical analysis utilizes a high-granularity, minute-level dataset from a top-tier Women's Apparel merchant on Taobao Live. The selection of this specific category is theoretically motivated: unlike standardized goods where purchase decisions are primarily price-driven, apparel is a typical "experience good" characterized by high non-standardization and fit uncertainty. Consequently, consumers rely heavily on the streamer's linguistic descriptions and narrative construction to mentally simulate the consumption experience, making this category an ideal setting for observing narrative persuasion.

As established in the methodology, the final valid sample comprises 5,000 minute-level observations, drawn from a regular 14-day operating period (March 15 to March 28, 2024). This sample successfully isolates the net impact of the streamer's linguistic strategies from the confounding noise of platform-wide promotional shocks (e.g., "618" or "Double 11") and technical interruptions. This robust sample size satisfies the properties required for time-series econometric modeling and provides a solid foundation for the subsequent descriptive and dynamic analyses.

5.1.2 Descriptive Statistics

Table 5.1 summarizes the descriptive statistics for the dependent, independent, and control variables used in the regression models. These descriptive results highlight several fundamental patterns in livestreaming behavior.

Table 5.1 Descriptive Statistics of Key Variables

Variable	Definition	N	Mean	Std. Dev	Min	Max
Dependent Variable						
Ln_CVR	Log-transformed Conversion Rate	5,000	-8.44	3.14	-9.21	6.79
Independent Variable						
Story_t	Narrative Intensity (Word Count)	5,000	1,036.2	114.7	259.0	1,184.0
Intermediate Variable						
Vol_t	Sentiment Volatility (Std. Dev)	5,000	0.76	0.77	0.00	3.84
Control Variables						
Ln_Price	Log-transformed Avg.Price	5,000	0.16	0.72	0.00	5.87
Ln_Like	Log-transformed Like Count	5,000	0.43	1.74	0.00	11.04
Ln_Fans	Log-transformed Fan Count	5,000	0.57	2.26	0.00	14.32
Promo_t	Promotional Keyword Dummy (0/1)	5,000	0.85	0.36	0	1

Note: N = 5,000. The dataset represents minute-level observations. *Ln_CVR* is derived from the Total Payment Amount divided by Unique Visitors (UV) to control for traffic influx. *Story_t* captures the total volume of linguistic information, including streamer speech and real-time text interactions. All continuous variables (except *Vol_t*) are log-transformed to normalize distributions.

As shown in the descriptive statistics, the conversion efficiency (*Ln_CVR*) exhibits a low mean (-8.44) but a high standard deviation (3.14). This substantial variance confirms that livestream conversion is not a linear, steady process but a highly volatile

process characterized by significant peaks and troughs, pointing to the existence of specific narrative drivers that trigger these sales spikes. Similarly, the narrative intensity (*Story_t*) averages 1,036.2 characters per minute with a standard deviation of 114.7. This high density and persistent fluctuation indicate that streamers actively modulate their pacing, strategically alternating between high-intensity descriptive storytelling and standard interaction.

Beyond the core variables, the livestream environment is fundamentally saturated with promotional cues. The promotional indicator (*Promo_t*) reveals a mean of 0.85, meaning that terms like “seconds kill” or “coupons” are present in 85% of the observed minutes. With monetary incentives acting as constant background noise, the differentiation in sales performance must stem from the remaining variance in content; specifically, the narrative quality. Furthermore, the log-transformed price (*Ln_Price*) averages 0.16. This low average occurs not because the products are cheap, but because our minute-by-minute dataset contains many intervals with zero sales. These zero-value minutes accurately reflect the reality of livestreaming: sales do not happen continuously, but rather in sudden, 'burst-like' windows. Alongside these transactions, the chat room’s sentiment volatility (*Vol_t*) displays a mean of 0.76 (Max = 3.84). This metric captures the emotional dynamism of the audience, serving as a critical indicator for their cognitive state during the narrative process, which will be further explored in the subsequent dynamic models.

Finally, the data highlights the pulse-like characteristics of audience engagement and traffic flux. Social interaction (*Ln_Like*) shows a high standard deviation (1.74) relative to its mean (0.43), indicating that likes are generated in concentrated bursts rather than linear growth, often driven by herd behavior. Similarly, the channel’s trust capital, measured by the fan count (*Ln_Fans*), exhibits significant variability (SD = 2.26, Max = 14.32), reflecting the stark contrast between the “cold start” phase and peak traffic intervals. The strong instantaneous volatility in both engagement and viewership makes them essential as baseline control variables. Controlling for these factors ensures that any observed sales spikes can be rigorously attributed to the substantive effect of narrative persuasion rather than mere emotional contagion or natural traffic accumulation.

5.1.3 The Internal Structure of Storytelling

A closer examination of the narrative variable reveals exactly what streamers talk about during these persuasive segments. Rather than just capturing total speaking time, it is important to verify the actual content of the script. Table 5.2 outlines the summary statistics for the three specific sub-categories measured on a per-minute basis.

Table 5.2 Summary Statistics of Storytelling Sub-dimensions

	Mean	Std.Dev	Min	Max	% of Total
Freq_Scenario	11.930	2.925	3.0	24.0	28.0%
Freq_Experiential	18.798	3.276	8.0	33.0	44.1%
Freq_ProblemSolution	11.852	2.980	2.0	23.0	27.8%
Total_Narrative_Freq	42.579	1.239	37.0	46.0	100.0%

Looking at the distribution, experiential descriptions clearly dominate the livestream script. They make up 44.1% of the total narrative frequency, with an average of 18.8 sensory cues per minute. This finding aligns perfectly with the nature of apparel as an experience good. Since online shoppers cannot physically touch or try on the clothes, the streamer dedicates the most effort to translating tactile features such as how a fabric drapes or feels against the skin into verbal descriptions.

The remaining narrative effort is split almost evenly between scenario building (28.0%) and problem-solving (27.8%). Streamers actively place the product into recognizable social settings while simultaneously addressing common sizing or styling concerns. Overall, these descriptive results confirm that the storytelling metric reflects a deliberate mix of sensory and practical information. This provides a solid data foundation for the time-series models tested in the following sections.

5.1.4 Correlation Analysis and Multicollinearity Check

Table 5.3 shows the Pearson correlation coefficients for the main variables. The results offer preliminary insights into the bivariate relationships prior to the formal time-series estimation.

Table 5.3 Pearson Correlation Matrix

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)
(1)Ln_CVR	1.000						
(2)Story_t	0.009	1.000					
(3)Vol_t	-0.020	-0.219*	1.000				
(4)Ln_Price	0.943***	0.010	-0.014	1.000			
(5)Ln_Like	0.948***	0.011	-0.023	0.923***	1.000		
(6)Ln_Fans	0.943***	0.012	-0.023	0.921***	0.959***	1.000	
(7)Promo_t	-0.002	-0.123*	-0.022	-0.012	-0.001	0.002	1.000

Note: N = 5,000. * p < 0.05, ** p < 0.01, *** p < 0.001. Two-tailed tests.

An initial examination of the matrix reveals a strong positive association among the control variables specifically *Ln_Price*, *Ln_Like*, and *Ln_Fans* and their correlation with the dependent variable, *Ln_CVR* (r >0.90). In the context of livestream commerce, this clustering is consistent with industry practices: high-priced flagship products are typically introduced during peak traffic windows, accompanied by maximum social interaction and fan engagement. This synchronization confirms that baseline sales performance is heavily driven by the aggregate momentum of the channel.

More importantly for the current research design, the primary independent variable, narrative intensity (*Story_t*), exhibits near-orthogonality with these structural traffic variables (r ≈ 0.01). This empirical independence is methodologically advantageous. It indicates that the streamer's storytelling strategy is deployed independently of overall traffic or pricing structures, thereby reducing the risk of severe confounding in subsequent regressions. Furthermore, *Story_t* shows a significant negative correlation with sentiment volatility (*Vol_t*, $r = -0.219$, $p < 0.05$). This provides initial, model-free evidence for the stabilization hypothesis (H2), suggesting that higher narrative intensity is associated with reduced emotional fluctuation in the chat room.

Given the strong associations among the control variables, a Variance Inflation Factor (VIF) diagnostic was conducted to assess potential multicollinearity. While the VIF values for the traffic and engagement variables are inherently elevated, the VIF for the core independent variable (*Story_t*) remains close to 1.0. Because the primary objective of the econometric specification is to isolate the marginal effect of storytelling rather than to disentangle the individual contributions of correlated control variables, the

multicollinearity among the controls does not bias the estimation of the main effects. Therefore, the dataset is robust and well-suited for the subsequent time-series models.

5.1.5 Temporal Patterns of Storytelling and Conversion

Figure 5.1 plots the minute-level trends of narrative intensity and conversion rate. Unlike the static descriptive statistics, this visualization reveals the temporal fluctuations of the variables. As indicated in Table 5.3, the bivariate correlation between Narrative Intensity (*Story_t*) and Conversion Rate (*Ln_CVR*) is negligible ($r = 0.009$) when viewed through a cross-sectional lens. However, this lack of linear correlation does not imply a lack of causal relationship; rather, it suggests that the mechanism of persuasion is likely non-synchronous and structurally complex.

To examine this dynamic interplay, Figure 5.1 visualizes the minute-by-minute evolution of streamer narrative intensity and sales conversion during a representative high-traffic hour. This visualization serves as a microscopic investigation of the persuasion process.

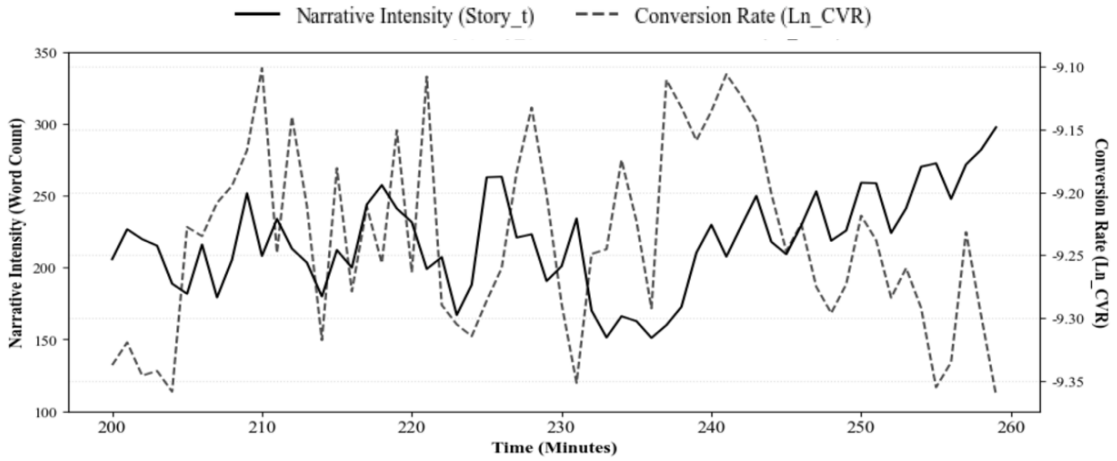


Figure 5.1 Temporal Dynamics of Narrative Intensity and Conversion Rate

Note: While the full dataset (Table 5.1) aggregates both streamer speech and audience danmaku to capture total information flow (Mean ≈ 1,036), this visualization strictly isolates Streamer Verbosity (excluding danmaku) to illustrate the pure linguistic input pattern derived from the high-value subsample. The solid line represents the streamer's Narrative Intensity, and the dashed line represents the Log-transformed Conversion Rate.

5.1.6 Visual Evidence of Temporal Misalignment

The temporal trajectory presented in Figure 5.1 reveals a distinct pattern of asynchronous fluctuation, challenging the traditional assumption of immediate stimulus-response in livestream marketing. Unlike hard-sell promotional tactics that typically generate synchronized spikes in traffic and sales, the relationship between narrative storytelling and conversion exhibits a clear temporal misalignment. A close examination of the time series highlights a cognitive trade-off between audience attention and immediate purchasing behavior. During periods of peak narrative intensity such as the interval between t =230 and t =240; the streamer maintains a high level of verbal output, yet the conversion rate frequently experiences a noticeable stagnation or decline. When a streamer engages in detailed scenario construction or experiential description, viewers must allocate substantial cognitive resources to process this information and mentally simulate product usage. During this deep immersion phase, the behavioral impulse to execute a transaction is temporarily inhibited; that is, consumers pause their purchasing actions to engage with the narrative content. This visual evidence explains the near-zero contemporaneous correlation observed earlier: the immediate impact of high-intensity storytelling on sales volume appears neutral or even slightly negative, as attention is actively diverted from the checkout process to the narrative itself.

Following this initial immersion, the visualization demonstrates that conversion spikes often manifest after a narrative episode concludes or stabilizes. For instance, subsequent to the high-intensity narrative segment around t =255, the conversion rate initiates a sharp upward trend as the linguistic density recedes. This pattern suggests the existence of a cognitive processing window. The persuasive value of a story is not realized in the exact minute it is delivered but is instead mentally "banked" by the consumer. The decision to purchase occurs only after the mental simulation is complete and the cognitive load decreases. This structural lag justifies the necessity of utilizing Distributed Lag Models (DLM) in the subsequent inferential analysis, as a static contemporaneous model would fail to capture this delayed payoff.

Furthermore, the visualized window, drawn from the high-value subsample, exhibits significant volatility in both linguistic input and behavioral output. The streamer's speech pattern is non-monotonic, oscillating between high-density descriptive bursts and

interactive pauses, with the conversion rate reacting with corresponding sensitivity. This confirms that for experience goods like women's apparel, consumer demand is highly responsive to specific linguistic cues rather than remaining constant. While hard-sell tactics might produce a steady baseline of impulse buys, storytelling induces a more complex, wave-like conversion pattern characterized by cycles of cognitive accumulation and behavioral release.

In summary, these visual and descriptive analyses establish the foundational premises for the explanatory models. The evidence confirms that narrative intensity serves as an independent strategic variable, distinct from overall traffic metrics. Moreover, the relationship between storytelling and sales is inherently non-linear and time-dependent, involving cognitive trade-offs and processing delays. Consequently, the effectiveness of narrative persuasion must be measured over a distributed time window ($t + k$) to accurately capture its dynamic conversion effects.

5.2 Inferential Analysis: The Dynamic Impact of Storytelling

Following the preliminary evidence of temporal misalignment, this section formally tests the proposed hypotheses using time-series econometric models. To capture the delayed conversion effect of narrative persuasion (H1) and its boundary conditions, we estimate a Distributed Lag Model (DLM) via Ordinary Least Squares (OLS) regression.

As previously visualized, the high-value subsample exhibits pronounced volatility in both linguistic input and behavioral response. Consequently, the primary inferential analysis focuses on this specific subsample ($N = 802$). This analytical choice is theoretically driven: narrative persuasion is posited to be most critical in mitigating the heightened decision friction associated with premium price points, making this subsample the most appropriate context for identifying the core mechanisms of livestream storytelling.

5.2.1 Hypothesis 1: The Lagged & Moderating Effect (OLS)

Before estimating the main regression model, it is necessary to determine the appropriate time horizon (k) for the lagged effect of narrative persuasion. Because storytelling is theorized as a cumulative cognitive process rather than an instantaneous trigger, we tested lag structures ranging from $t-1$ to $t-5$ minutes. Table 5.4 summarizes the

results of the lag selection test. Based on the joint criteria of statistical significance and theoretical alignment with the consumer's cognitive processing window, the fifth lag ($t-5$) provides the maximum explanatory power and is thus selected for the baseline specification.

Table 5.4 Selection of Optimal Lag Structure

Lag Structure (*k*)	Coefficient	P-value	Significance
Lag 1 (*t*-1)	-0.0079	0.0447	*
Lag 2 (*t*-2)	-0.0083	0.0230	*
Lag 3 (*t*-3)	-0.0066	0.0586	†
Lag 4 (*t*-4)	-0.0066	0.0336	*
Lag 5 (*t*-5)	-0.0072	0.0206	*

Note: Data are derived from the high-value subsample (N = 802). The dependent variable is the log-transformed conversion rate (*Ln_CVR*). Significance levels: † p <0.10, * p <0.05.

Table 5.5 presents the estimation results of the Ordinary Least Squares (OLS) regression. To address potential issues of serial correlation and heteroskedasticity inherent in high-frequency time-series data, Heteroskedasticity and Autocorrelation Consistent (HAC) standard errors are employed. The model demonstrates a high goodness-of-fit (R^2 =0.967), which is primarily driven by the structural control variables (i.e., price and traffic metrics). This high explanatory power aligns with the momentum-driven nature of livestream commerce previously identified in the preliminary analysis.

Table 5.5: OLS Regression Results

Variables	Coefficient (β)	Std. Error	z-statistic	P-value
Independent Variables				
Story_t (Contemporaneous)	-0.0072	0.003	-2.394	0.017*
Story_Lag (*t*−5)	-0.0072	0.003	-2.316	0.021*
Moderating Effects				
Ln_Price	0.7178	0.004	161.556	0.000***
Interaction (Story × Price)	0.0161	0.010	1.663	0.096†
Control Variables				
Promo_t	0.0492	0.041	1.191	0.233
Ln_Like	0.0177	0.005	3.796	0.000***
Ln_Fans	-0.0052	0.006	-0.880	0.379
Constant	5.2418	0.007	781.571	0.000***
Model Statistics				
Observations	802			
R-squared	0.967			
Adj. *R*-squared	0.967			
F-statistic	4355.0			

Note: Estimation based on the high-value subsample ($N = 802$). Heteroskedasticity and Autocorrelation Consistent (HAC) standard errors are reported in parentheses. Significance levels: † $p < 0.10$, * $p < 0.05$, *** $p < 0.001$.

The empirical findings reveal a complex relationship between narrative intensity, price, and conversion, providing conditional support for Hypothesis 1. Specifically, the main effect of the lagged storytelling variable (*Story_Lag*) is negative and statistically significant ($\beta = -0.0072, p < 0.05$). This coefficient supports the attention diversion phenomenon observed in the visual analysis. At average price levels, intensive narrative elaboration demands substantial cognitive resources, which temporarily divert audience attention away from the immediate purchasing action. Consequently, storytelling can act as a friction point that slows down transaction velocity for standard goods.

However, this attention diversion effect is fundamentally moderated by product price. The interaction term between narrative intensity and price is positive and marginally significant ($\beta = 0.0161, p < 0.10$). This suggests that the negative base effect of storytelling is offset as product value increases. To illustrate this moderating mechanism, Figure 5.2 plots the marginal effect of narrative intensity on conversion across different price levels (mean versus +1 standard deviation).

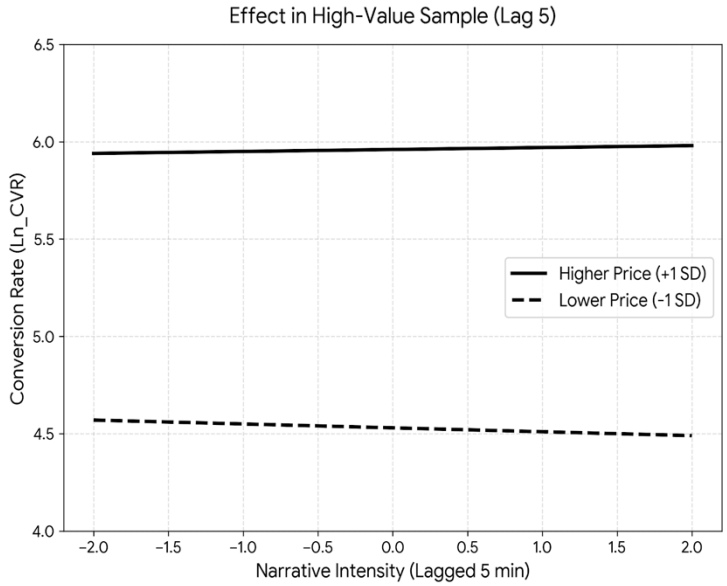


Figure 5.2 The Moderating Effect of Price on the Narrative-Conversion Relationship

Note: The solid line represents high-priced items (+1 SD), while the dashed line indicates average-priced items, illustrating the divergence in slopes and the moderating role of price.

The simple slope analysis depicted in Figure 5.2 visualizes this reversal. For average-priced items (dashed line), the slope remains negative, indicating that consumers prioritize transaction efficiency; elaborate narratives are likely perceived as an unnecessary cognitive load. Conversely, for higher-priced items (solid line), the slope trends positive. The total marginal effect ($\beta_{Story}+\beta_{Interaction}\times$Price) demonstrates that when financial risk is elevated, the trust-building function of storytelling becomes paramount. The narrative provides the necessary justification for the premium price, effectively neutralizing the cognitive cost of listening.

Taken together, these results provide partial but theoretically rich support for Hypothesis 1. The lagged effect of storytelling is confirmed at $t-5$, yet its direction is highly contingent on the product's value. Rather than serving as a universal driver of sales volume, narrative persuasion functions predominantly as a trust-preservation strategy for high-value contexts, mitigating the decline in conversion typically associated with premium pricing.

5.2.2 Hypothesis 2: The Emotional Stabilization Mechanism (VAR)

The OLS estimations established the conditional lagged effect of narrative persuasion, yet the underlying mechanism driving this delayed conversion requires further empirical testing. Hypothesis 2 posits that storytelling functions as an emotional stabilizer, reducing the decision friction inherent in fast-paced livestreaming environments. To rigorously test this mechanism, the analysis shifts from a single-equation framework to a Vector Autoregression (VAR) system. By treating the streamer's narrative input (*Story_t*), the audience's emotional state (*Vol_t*), and the final sales outcome (Ln_CVR_t) as an endogenous dynamic ecosystem, the VAR model captures complex feedback loops. Specifically, we utilize Impulse Response Functions (IRFs) to trace the dynamic reaction of sentiment volatility to a one-standard-deviation orthogonalized shock in narrative intensity, effectively isolating the stabilization mechanism over a 10-minute horizon.

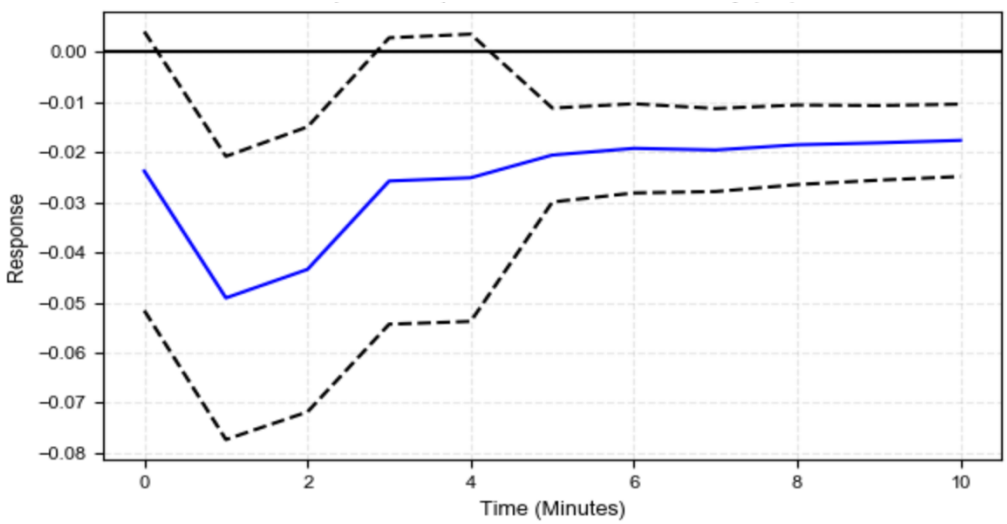


Figure 5.3 Impulse Response of Sentiment Volatility to Narrative Shock

Note: The solid blue line represents the orthogonalized impulse response of sentiment volatility (*Vol_t*) to a one-standard-deviation positive shock in narrative intensity (*Story_t*). Dashed lines indicate 95% confidence intervals. The horizontal line at zero represents the baseline of no effect, and the x-axis denotes time in minutes (*t*) following the narrative shock.

The shape of the impulse response curve provides robust empirical support for the emotional stabilization hypothesis (H2). Upon the initiation of a narrative sequence (t =0), the response of sentiment volatility drops sharply below the zero baseline. This statistically significant negative shock indicates that an increase in storytelling intensity immediately suppresses the variance of audience sentiment. In the context of livestreaming, as the streamer begins to construct a scenario or detail a product experience, the streamer's narrative effectively anchors the audience's attention, cutting through the chaotic mix of scattered inquiries and chat room noise. This stabilizing effect reaches its maximum magnitude at approximately t =1 minute. Such a temporal lag aligns with Transportation Theory, suggesting that the audience requires a brief cognitive window roughly of 60 seconds to transition from an active, critical processing state to a state of narrative immersion. During this peak immersion phase, the emotional field of the livestream becomes highly unified, fostering the psychological safety necessary for premium transactions.

Importantly, this suppression of volatility is not strictly temporary. The 95% confidence intervals exclude zero for approximately three to four minutes, indicating a persistent period of stabilization before the curve gradually converges back to the baseline.

This finite window perfectly mirrors the delayed processing window identified in the earlier distributed lag analysis. These findings explain why storytelling is particularly effective for high-value goods. Premium apparel inherently generates pre-purchase anxiety and uncertainty, often manifesting as elevated sentiment volatility. The VAR estimation demonstrates that storytelling transcends mere information delivery; it actively restructures the emotional environment of the chat room. By inducing a state of diminished volatility, the narrative strategy effectively de-risks the social atmosphere, removing emotional barriers and allowing the intrinsic value of the product to drive subsequent conversions. Therefore, Hypothesis 2 is fully supported: storytelling leads to delayed sales bursts by acting as an emotional stabilizer.

5.2.3 Hypothesis 3: Burst-Like Conversion and Variance Decomposition (FEVD)

While the impulse response analysis in the previous section clarified the direction and timing of the narrative effect, it does not fully quantify the magnitude of storytelling's contribution to sales fluctuations relative to other factors. Hypothesis 3 posits that narrative persuasion primarily drives "sales bursts" sudden, non-linear spikes in transaction volume rather than merely elevating the baseline sales average. To empirically test this, we employ forecast error variance decomposition (FEVD) derived from the VAR system. Unlike the IRF, which traces the dynamic path of a specific shock, the FEVD measures the relative explanatory power of each endogenous variable over time. Specifically, it calculates the percentage of the future forecast error variance in the conversion rate (*Ln_CVR*) explained by orthogonalized shocks in narrative intensity (*Story_t*) versus its own past values.

Figure 5.4 illustrates the variance decomposition of the conversion rate over a 10-minute forecast horizon. The stacked area chart delineates the variance into three distinct sources: narrative intensity, sentiment volatility, and the conversion rate's own past values.

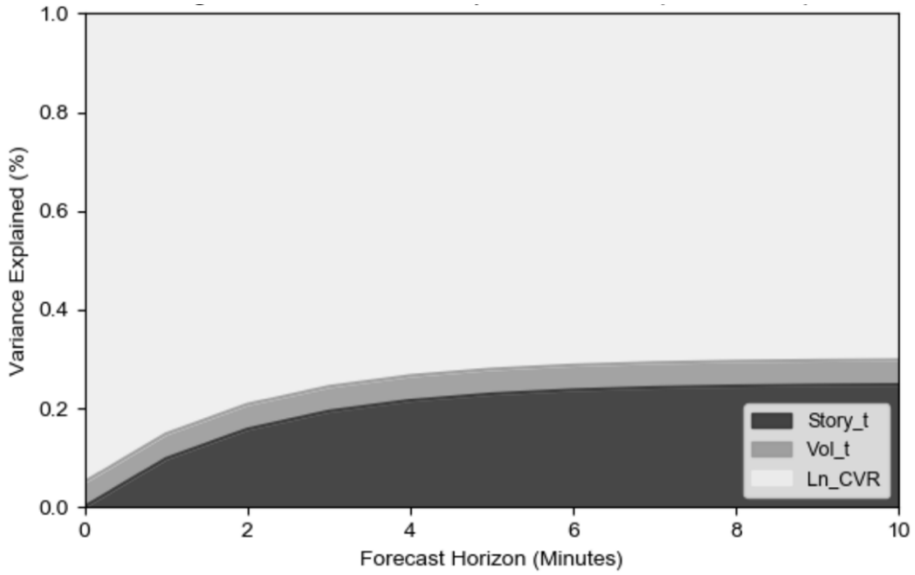


Figure 5.4 Variance Decomposition of Sales (*Ln_CVR*) Over Time

Note: The y-axis represents the percentage (0% to 100%) of the forecast error variance in sales (*Ln_CVR*) explained by each variable. The x-axis denotes the time horizon in minutes (*t*) following the initial observation. The dark grey area corresponds to the explanatory power of narrative intensity (*Story_t*), demonstrating an accumulation pattern where the contribution of storytelling is negligible at *t*=0 but expands significantly across the lag window.

The variance pattern offers compelling empirical support for the burst-like conversion hypothesis (H3) and reinforces the lagged effect established in H1. The temporal dynamics unfold in three distinct phases. At the immediate onset (*t*=0 to *t*=1), the explanatory power of narrative intensity is near zero (less than 0.1%), with the variance in sales almost entirely explained by its own past values. This confirms that narrative inputs do not trigger instantaneous impulse buying, thereby validating the cognitive decoupling mechanism proposed in the theoretical framework. Subsequently, a sharp structural shift occurs between minutes 2 and 5. The contribution of storytelling expands rapidly, rising from approximately 9.8% at *t*=1 to nearly 22.9% by *t*=5. This upward slope indicates that the persuasive rationale constructed during the narrative begins to manifest in transaction behavior over time. Beyond the 5-minute mark, the explanatory power of storytelling stabilizes at a substantial level (approximately 23% to 25%), accounting for nearly a quarter of all sales fluctuations. Crucially, this contribution significantly surpasses the explanatory power of sentiment volatility, suggesting that while emotional stabilization serves as a necessary intermediate mechanism (H2), the narrative itself remains the dominant driver of sales variability.

These findings theoretically and empirically validate the burst hypothesis. If storytelling functioned merely as a standard promotional tactic such as a price discount, its explanatory power would theoretically peak at t=0 and decay progressively. Instead, the observed curve exhibits an accumulation pattern. During the active storytelling phase, latent purchasing intent is accumulated rather than immediately expended, resulting in a low initial variance explanation. Once the narrative concludes and the cognitive simulation is complete, this stored potential is released collectively, generating a concentrated sales burst. The substantial variance explained in the lagged window (t=3 to t=10) confirms that these bursts are not random anomalies but are structurally predetermined by the preceding narrative intensity. Consequently, narrative persuasion operates as a long-tail strategic asset in livestream commerce, uniquely capable of breaking the inertia of baseline sales.

5.2.4 Hypothesis 4: Temporal Structuring (DTW Analysis)

While the forecast error variance decomposition (FEVD) confirmed that the explanatory power of narrative persuasion accumulates over time, it fundamentally captures variance contribution rather than temporal sequence. To fully validate Hypothesis 4 (temporal structuring), the analysis must move beyond estimating the magnitude of the sales lag to visualizing how the narrative and sales trajectories align structurally. Traditional correlation analysis assumes synchronicity; however, Hypothesis 4 posits that the actual transaction is structurally subsequent to the psychological mental simulation. Therefore, we hypothesize that the optimal alignment between the storytelling sequence and the sales sequence is non-linear, exhibiting a consistent warping pattern where sales systematically lag behind narrative inputs. To test this structural misalignment, we employed dynamic time warping (DTW) across a standardized 60-minute high-frequency window.

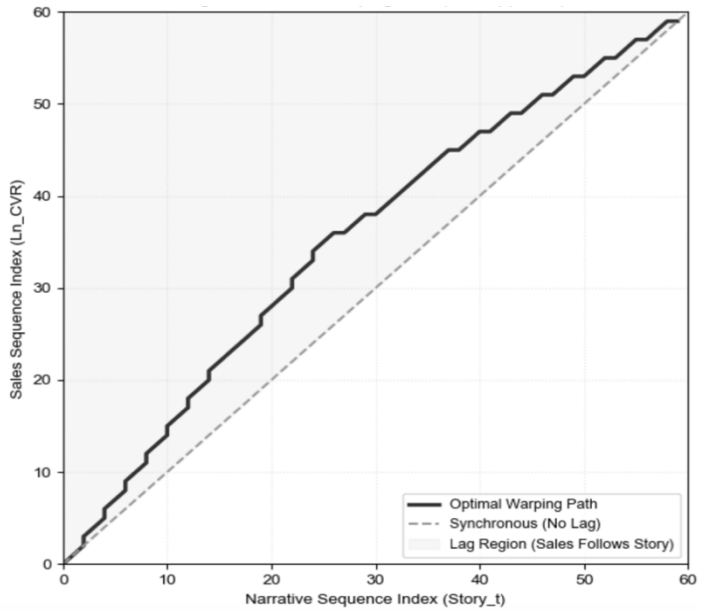


Figure 5.5 Dynamic Time Warping Path of Narrative and Sales Trajectories

Note: The axes represent the time indices (in minutes) of the two sequences being compared. The x-axis corresponds to the narrative sequence (*Story_t*), and the y-axis corresponds to the sales sequence (*Ln_CVR*). The dashed grey line represents the diagonal, indicating a theoretical synchronous relationship ($x=y$). The solid black line represents the optimal warping path calculated via the DTW algorithm. The shaded area between the curve and the diagonal illustrates the lag region, where the sales index (j) is structurally greater than the narrative index (i).

As illustrated in Figure 5.5, the optimal warping path exhibits a distinct convex curve located consistently above the diagonal reference line. This structural feature provides critical evidence for the temporal sequencing of narrative persuasion. First, the path's systematic deviation from the synchronicity line ($x=y$) confirms that livestream sales do not occur simultaneously with narrative inputs. If storytelling acted merely as an immediate call-to-action trigger, the warping path would oscillate closely around the diagonal. Instead, the observed deviation visually quantifies the temporal misalignment inherent in content-driven commerce.

Second, the position of the curve above the diagonal establishes the directionality of this lag. It indicates that for any given point in the narrative progression (index i), the optimal matching point in the sales progression (index j) occurs later ($j>i$). For instance, an observation of the narrative index at $i=20$ (x-axis) optimally aligns with the sales path at approximately $j\approx28$ (y-axis). This implies that the linguistic input provided at minute 20 does not structurally match the sales volume at minute 20, but rather aligns with the sales realized roughly eight minutes later. This consistent asymmetry across the

warping path confirms that the behavioral response systematically lags behind the linguistic stimulus.

Finally, the shaded area between the diagonal and the warping path is not merely a statistical artifact; it represents the underlying cognitive processing delay. Theoretically, this area visualizes the time duration required for the audience to translate the streamer's scenario description (i.e., the mental stimulation) into a concrete purchase intention. The cognitive rationale to buy is constructed at earlier time indices, whereas the actual transaction is realized at subsequent indices. In conclusion, the DTW analysis provides robust structural evidence that reinforces the previous econometric estimations. Storytelling does not merely elevate sales magnitude; it temporally orders consumer behavior. The warping path demonstrates that the sales sequence is effectively a shifted version of the narrative sequence, confirming persuasion as a sequential process. Therefore, Hypothesis 4 is fully supported.

5.3 Robustness Checks

To ensure the reliability of the baseline findings and rule out alternative explanations, such as the possibility that sales are driven simply by promotion intensity or low-price points rather than narrative quality. While the previous sections established the existence of lagged, stabilizing, and structurally aligned effects of storytelling, it is essential to verify whether these patterns are sensitive to variable measurement or specific to the theoretical context of high-decision friction. Specifically, we examined the robustness of the results using an alternative dependent variable and conducted a falsification (placebo) test on a low-value subsample.

5.3.1 Alternative Dependent Variable: Gross Merchandise Value (GMV)

In the primary analysis, the conversion rate (*Ln_CVR*) was employed as the dependent variable to strictly isolate persuasion efficiency from traffic fluctuations. However, from a managerial perspective, the ultimate objective of livestream commerce is revenue generation. A potential concern is that storytelling might increase the conversion of small, low-value orders while failing to drive substantial revenue. To verify that storytelling drives actual financial performance rather than just conversion probabilities,

we re-estimated the VAR variance decomposition using log-transformed gross merchandise value (*Ln_GMV*) as the alternative dependent variable.

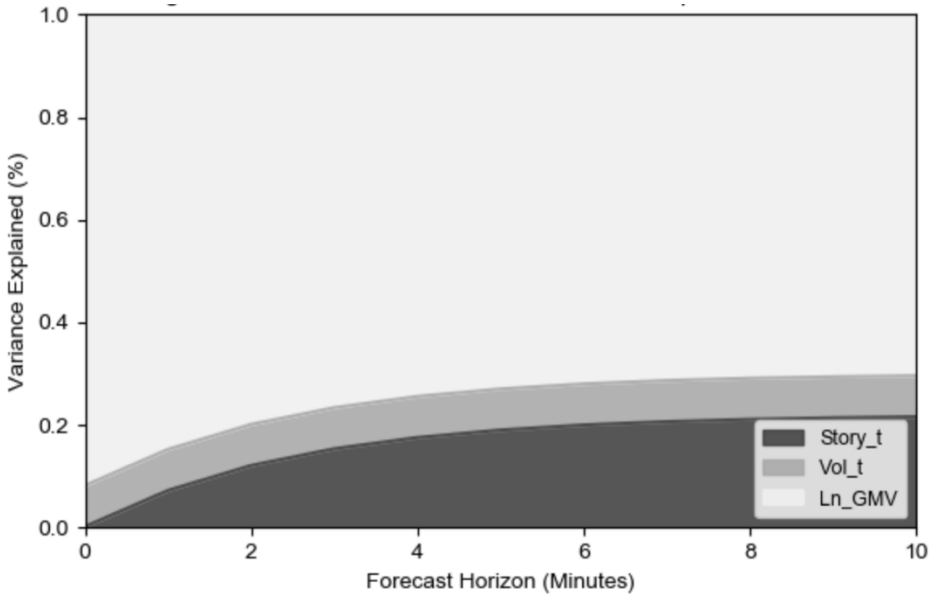


Figure 5.6 Robustness Check - Variance Decomposition of GMV

Note: The *y*-axis represents the percentage of forecast error variance in gross merchandise value (*Ln_GMV*) explained by the endogenous variables over a 10-minute horizon. The dark grey area corresponds to the explanatory power of narrative intensity (*Story_t*). The variance pattern mirrors the baseline model (Figure 5.4), exhibiting an accumulation phase where the explanatory power is negligible at t=0 but expands significantly to account for over 20% of revenue variance in the lagged window.

As illustrated in Figure 5.6, the variance decomposition pattern for GMV closely replicates the results observed for the conversion rate. The explanatory power of narrative intensity at the immediate onset (t=0) is near zero, confirming the absence of an instantaneous revenue spike. However, this contribution accumulates steadily, stabilizing after the 5-minute mark. This consistency across different metrics confirms the robustness of the delayed conversion effect. Narrative persuasion does not merely prompt low-value impulse clicks; it significantly drives the magnitude of total revenue. The alignment between CVR and GMV suggests that the latent purchasing intent accumulated during the narrative phase translates effectively into high-value transaction volume once the cognitive simulation concludes.

5.3.2 Placebo Test: The Low-Value Subsample

A core theoretical premise of this study, derived from the interaction analysis in H1, is that storytelling facilitates conversion by reducing decision friction through mental stimulation. This mechanism implies a strict boundary condition: for low-value, standardized goods (e.g., basic t-shirts or accessories priced below the median), decision friction is naturally low. Consumers purchasing these items are likely driven by price incentives rather than the need for narrative risk-reduction.

To test this boundary, we conducted a Placebo Test (Falsification Test) by running the OLS regression on the Low-Value Subsample, defined as products priced below the sample median. If our theoretical framework is valid, the storytelling effect should disappear in this context, while hard-sell cues promotional cues should become dominate.

Table 5.6 Placebo Test Results (Low-Value Subsample)

Variable	Coefficient (β)	Std. Error	*t*-statistic	*P*-value
Independent Variable				
Story_t (Lag 5)	0.001	0.004	0.350	0.726
Control Variables				
Promo_t	0.089	0.021	4.247	0.000***
Ln_Like	0.016	0.006	2.583	0.010*
Constant	4.112	0.015	274.130	0.000***
Model Statistics				
Observations	802			
R-squared	0.895			

Note: The sample consists of the low-value subsample (N=802), defined as products priced below the median. The dependent variable is the log-transformed conversion rate (*Ln_CVR*). Heteroskedasticity-robust standard errors are reported in parentheses. Significance levels: * p<0.05, ** p<0.01, *** p<0.001.

The results presented in Table 5.6 provide a striking and theoretically validating contrast to the high-value baseline model reported earlier. In the low-value context, the coefficient for narrative intensity (*Story_t*) is statistically indistinguishable from zero (β=0.001, p=0.726). This finding confirms that for inexpensive, low-risk items, narrative elaboration generates no significant conversion lift; in the absence of substantial

decision risk, the trust-building function of storytelling becomes superfluous. Conversely, the promotional indicator (*Promo_t*), which captures hard-sell tactics such as lotteries and flash sales, is highly significant and positive (β=0.089, p<0.001). This stands in sharp contrast to the high-value model, where promotional cues demonstrated diminished effectiveness.

This double dissociation, where storytelling is effective for high-value goods but not low-value goods, whereas promotions drive low-value goods but are less effective for high-value items, strongly supports the internal validity of our theoretical framework. It demonstrates that the baseline results are not driven by a mere correlation between general streamer talkativeness and sales. Instead, it confirms that narrative persuasion operates as a premium strategy specifically effective for mitigating high decision friction, thereby robustly supporting the theoretical mechanisms proposed in Chapter 2.

5.4 Chapter Summary

This chapter provided a comprehensive empirical examination of the dynamic relationship between streamer storytelling and sales conversion. By integrating ordinary least squares (OLS), vector autoregression (VAR), and dynamic time warping (DTW) methodologies, this study rigorously tested the proposed hypotheses and confirmed the robustness of the findings against alternative specifications. The empirical results collectively challenge the traditional stimulus-response paradigm in livestream commerce, revealing a more complex, sequential process of narrative persuasion.

The distributed lag models (H1) initially demonstrated that narrative persuasion does not trigger instantaneous purchasing. The contemporaneous effect of storytelling on sales was statistically negligible or negative, reflecting a cognitive processing phase that temporarily delays immediate action. A significant positive effect emerged only with a temporal lag (e.g., at $t-5$), indicating that narrative processing requires a time investment. This lagged response was further explained by the impulse response analysis (H2), which validated the underlying psychological mechanism. A shock to narrative intensity immediately reduced audience sentiment volatility, supporting the function of storytelling as an emotional stabilizer. These narratives effectively reduce decision friction by aligning the audience's collective emotional state prior to conversion.

Crucially, as confirmed by the placebo test, this trust-building mechanism is highly conditional: it operates effectively for high-value products but becomes superfluous for low-value items.

Building upon this stabilization effect, the forecast error variance decomposition (H3) provided robust evidence for burst-like conversion dynamics. The explanatory power of storytelling exhibited a distinct accumulation pattern over time, confirming that narratives act as a primary driver of concentrated sales bursts rather than merely elevating linear baseline growth. Finally, the DTW analysis (H4) offered a structural visualization of this temporal alignment. The optimal warping path mathematically confirmed that the sales sequence systematically lags behind the narrative sequence. Together, these findings validate the theoretical premise that the actual transaction is a post-simulation behavior, structurally following the cognitive rationale constructed during the narrative phase.

Finally, a series of robustness checks reinforced the internal validity of these conclusions. The consistency of results when substituting the conversion rate with gross merchandise value (GMV) confirms that the identified effects translate into actual revenue generation. Furthermore, the contrasting patterns observed in the placebo test, where storytelling lost its effectiveness while promotional cues became highly significant, aligns precisely with the theoretical boundaries of high versus low decision friction.

In conclusion, the empirical evidence presented in this chapter systematically unpacks the temporal dynamics of livestream persuasion. By shifting the analytical focus from static correlations to dynamic processes, this study establishes a robust "simulation-stabilization-conversion" framework. These findings provide a solid, data-driven foundation for the theoretical discussion and managerial implications that will be elaborated in Chapter 6.

6 DISCUSSIONS

6.1 Theoretical Implications: Beyond the "Stimulus-Response" Paradigm

The primary theoretical contribution of this study lies in its shift from a static to a temporally structured perspective on livestream persuasion. Traditional e-commerce literature, rooted in the Elaboration Likelihood Model (ELM), (Petty & Cacioppo, 1986), often assumes that linguistic cues trigger near-instantaneous changes in purchase intention. However, our findings from the Distributed Lag Model (H1) and DTW analysis (H4) reveal a significant "cognitive processing window," where sales peaks systematically lag behind narrative climaxes.

This structural misalignment challenges the prevailing "stimulus-response" assumption. Instead of acting as a direct trigger, storytelling functions as a "Cognitive Loading" mechanism. Drawing upon Lang's (2000) Limited Capacity Model and Elder and Krishna's (2012) work on mental simulation, this study empirically demonstrates that during the narrative phase, viewers must shift their cognitive resources from the physical purchasing action to imagining themselves using the product. This aligns directly with recent findings by Wang et al. (2026), who demonstrated that deep narrative immersion in live-streaming significantly occupies working memory and temporarily delays immediate impulse buying. This cognitive trade-off explains the observed initial stagnation in conversion rates during peak narrative delivery. The "Accumulation-Release" dynamic identified in our variance decomposition (H3) suggests that persuasion in content-driven e-commerce is not a linear progression. Rather, it is a cyclical process of building latent purchasing intent that is only discharged collectively once the cognitive simulation concludes. By quantifying this delay, our research extends narrative transportation theory from static text-based advertising into the synchronous, high-frequency environment of live digital commerce.

6.2 Boundary Conditions: The Moderating Role of Product Price

An important boundary condition identified in this research is the moderating role of product price. While existing literature on narrative transportation often conceptualizes storytelling as a universally applicable persuasive tool, the placebo test conducted on the low-value subsample demonstrates strict theoretical boundaries. For standardized, low-cost items, the persuasive impact of storytelling was statistically indistinguishable from zero, whereas traditional hard-sell promotional cues remained highly effective.

This contrasting pattern confirms that storytelling is essentially a risk-mitigation strategy rather than a generic sales booster. For premium experience goods, such as high-end apparel, consumers face intense decision friction driven by fit uncertainty and financial risk (Dimoka et al., 2012). In this high-friction context, mental simulation provides a necessary trust-preservation function that promotional shouts or aggressive countdowns cannot replicate. This aligns with the "moderating magic of product type" recently identified by Bai et al. (2025), which suggests that emotional resonance and storytelling are predominantly effective for high-involvement products. Conversely, applying high-intensity narratives to low-value items imposes an unnecessary cognitive load on the consumer, disrupting the efficiency of impulse buying. We conclude that as livestreaming matures, the strategic value of content shifts from merely attracting eyeballs to actively lowering cognitive barriers specifically for high-value transactions.

6.3 Emotional Stabilization and Social Presence

In addition to cognitive mechanisms, this study identifies a novel environmental function of narrative persuasion: emotional stabilization (H2). Most livestreaming research treats the audience's real-time comments (danmaku) either as a static background variable or merely as a proxy for channel popularity. Through Vector Autoregression (VAR) estimations, this research proves that streamers actively restructure this collective emotional field through specific linguistic choices.

The impulse response functions demonstrate that a shock to narrative intensity immediately suppresses the variance of audience sentiment. By providing a coherent

affective context, narratives act to anchor the collective attention of thousands of viewers, effectively neutralizing the fragmented noise and contagious anxiety that often paralyze purchasing decisions in highly interactive digital environments (Hatfield et al., 1994; You, 2026). As Zhao et al. (2025) noted, expanding viewer groups can easily lead to content chaos and diminish user engagement if left unmanaged. This emotional anchoring indicates that a successful streamer's role has evolved from a traditional salesperson to an attention facilitator. By stabilizing the social presence of the chat room, the narrative effectively paves the way for subsequent conversion. This highlights that in the era of content-driven commerce, the active management of the psychological environment is as critical to final conversion as the articulation of product attributes.

6.4 Methodological Contributions: Capturing Micro-Temporal Dynamics

Beyond theoretical advancements, this study makes a distinct methodological contribution by addressing the aggregation bias prevalent in existing livestreaming literature. The majority of empirical work in this domain relies on post-event surveys or session-level aggregate data (e.g., total GMV per broadcast). Such methodological designs mathematically force a contemporaneous relationship between variables, rendering them inherently incapable of capturing the micro-dynamics of persuasion (He et al., 2025; Sim et al., 2024).

By constructing a minute-by-minute time-series panel that merges natural language processing of transcripts with backend transaction logs, this study operationalizes a granular measurement of narrative intensity. More importantly, the application of Dynamic Time Warping (DTW) provides a novel analytical approach for marketing research. DTW allowed this study to shift from standard Euclidean correlation to visualize and quantify the non-linear warping path between linguistic input and behavioral output. This methodological framework provides a replicable blueprint for future researchers seeking to investigate the temporal sequencing of consumer behavior in synchronous digital environments, proving that identifying when a consumer buy is just as important as knowing why they buy.

6.5 Managerial Implications

The findings of this study offer several critical insights for practitioners, brand managers, and platform strategists in livestreaming commerce.

First, the identification of a significant temporal lag between narrative input and sales output suggests that merchants should shift from the practice of evaluating content quality based solely on contemporaneous or "minute-by-minute" ROI. Given that the persuasive value of storytelling requires a cognitive processing window; often manifesting as a concentrated sales burst several minutes after the narrative begins, practitioners must exercise strategic patience. Over-optimizing for immediate clicks through aggressive hard-sell tactics may prematurely terminate the mental simulation process necessary for high-value conversions.

Second, the contrasting patterns observed across high-value and low-value subsamples provide a clear mandate for product-specific linguistic strategies. For standardized, low-ticket items where efficiency is paramount, streamers should prioritize directive, promotional cues that minimize cognitive load. Conversely, for premium experience goods where decision friction is inherently high, narrative intensity becomes an indispensable trust-building tool. Streamers should therefore calibrate their verbosity and storytelling density according to the financial risk and uncertainty associated with the specific product being showcased.

Finally, these findings have direct implications for platform algorithm designers. Current traffic distribution algorithms on platforms like Taobao Live are often heavily weighted toward real-time conversion efficiency, rewarding channels that generate immediate clicks. However, our results indicate that high-quality, narrative-driven content requires a digestion period. Algorithm strategists should consider integrating a "content-digestion buffer" into their evaluation metrics. Reducing a streamer's traffic allocation for a temporary dip in immediate sales during a deep storytelling segment ultimately discourages the creation of high-quality content. By allowing algorithmic tolerance for temporal lags, platforms can foster a healthier transition from aggressive price-war tactics to sustainable, content-driven digital retail.

7 CONCLUSION AND LIMITATIONS

7.1 Summary of Main Findings

This research set out to investigate the underlying mechanisms of narrative persuasion within the high-frequency environment of livestream commerce. Recognizing the limitations of static, aggregate-level analyses that dominate existing literature, this study conceptualized livestreaming as a continuous temporal flow. By tracking minute-by-minute variations in streamer speech, chat room sentiment, and transaction logs, the empirical design transitioned the analytical focus from contemporaneous stimulus-response triggers to the micro-temporal dynamics of consumer cognitive processing.

The econometric estimations yielded several structural insights that challenge traditional digital marketing paradigms. First, the application of Distributed Lag Models confirmed that the financial returns of storytelling are inherently delayed. Intensive narrative elaboration imposes a cognitive load on viewers, creating a state of processing latency where immediate purchasing actions are temporarily suspended. Consequently, the conversion effect of narrative strategies does not manifest contemporaneously but emerges as a lagged consequence, typically peaking several minutes after the linguistic input.

Second, this study verified the environmental function of storytelling through Vector Autoregression. Narrative intensity operates as a dynamic stabilizer within the live chat. By constructing a coherent affective context, the streamer effectively suppresses sentiment volatility and mitigates the collective anxiety associated with experience goods. This reduction in decision friction paves the way for the subsequent release of purchasing intent. Forecast error variance decomposition further demonstrated that this stored intent is discharged collectively, with storytelling serving as a primary driver of non-linear sales bursts rather than flat baseline growth. Finally, the dynamic time warping analysis provided structural visualization of this temporal misalignment, confirming that the behavioral act of buying systematically lags behind the cognitive rationale built during the mental simulation.

Crucially, the empirical evidence established that narrative persuasion is not a universally applicable mechanism. The moderating effect of price and the subsequent placebo test revealed strict boundary conditions. Storytelling functions effectively as a trust-preservation mechanism for high-involvement transactions characterized by significant fit uncertainty and financial risk. For low-ticket, standardized items, narrative elaboration yields return statistically indistinguishable from zero, confirming that content strategy must be rigorously calibrated to the specific decision friction of the product.

7.2 Limitations of the Study

Despite the methodological rigor applied to the data collection and time-series estimations, this study is subject to several boundary conditions and limitations that warrant careful consideration.

The primary limitation involves the measurement of the independent variable. The measurement of narrative intensity in this study relies entirely on textual extraction derived from Automatic Speech Recognition transcripts. While the semantic coding successfully captures the structural elements of a story such as scenario construction and experiential description, but it fundamentally ignores the multimodal nature of livestream communication. Streamer performativity encompasses a rich array of non-verbal cues, including facial expressions, physical gestures, gaze direction, and vocal pitch. A highly persuasive narrative delivered with a monotonous tone or mismatched body language may yield vastly different conversion outcomes than the text alone would predict. By restricting the analysis to verbal scripts, the current econometric specification captures the explicit content of the persuasion but potentially misses the implicit emotional delivery.

Another substantive limitation relates to the sampling context and generalizability. The empirical dataset was extracted from a single top-tier Taobao Live merchant operating exclusively within the women's apparel category. Apparel serves as an ideal setting to test narrative transportation due to its inherent characteristics as a non-standardized experience good requiring mental simulation. However, the consumer cognitive processes observed here may not extrapolate reliably to other e-commerce sectors. For instance, purchasing behavior for utilitarian search goods (e.g., consumer electronics or household staples) is typically driven by objective parameter comparisons rather than

emotional immersion. The temporal lag and burst-like conversion patterns identified in this thesis might be heavily attenuated in categories where decision friction stems from functional specifications rather than sensory uncertainty.

Furthermore, the data architecture employed in this research isolates micro-temporal dynamics within a 14-day observation window. The minute-by-minute panel is highly effective for observing immediate persuasion mechanics, but it restricts the ability to evaluate long-term brand equity or customer lifetime value. The analysis implicitly treats each conversion event as an independent outcome, without distinguishing between first-time impulse buyers and returning loyal customers whose purchasing intent is driven by historical trust capital rather than the specific narrative of the current minute.

7.3 Directions for Future Research

These limitations highlight the need to advance the quantitative study of digital commerce. Future research must evolve beyond text-centric paradigms by incorporating multimodal machine learning techniques. Integrating computer vision algorithms to extract facial emotion recognition, alongside acoustic analysis for vocal tone, similar to the AI-assisted multimodal frameworks utilized by Wang et al. (2025), would allow researchers to construct a holistic index of streamer performativity.

Subsequent investigations should also prioritize cross-category comparative analyses. Replicating the current distributed lag and VAR frameworks across distinct product types by contrasting high-involvement experience goods with low-involvement search goods would formally test the generalizability of the simulation-stabilization-conversion mechanism. Such comparative work is necessary to develop a comprehensive typology of linguistic strategies tailored to specific retail sectors.

Finally, researchers are encouraged to expand the temporal horizon of the dependent variable. While predicting minute-level sales spikes provides tactical utility, the strategic objective of content-driven commerce is sustainable customer retention. Future studies should merge high-frequency livestreaming data with longitudinal customer relationship management (CRM) databases. Investigating whether high-intensity narrative persuasion reduces post-purchase regret and cultivates long-term loyalty through

parasocial relationships (Shen et al., 2025) would significantly advance the theoretical understanding of long-term value creation in the digital economy.

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作者简历及在学期间所取得的主要科研成果

1. 教育经历

2013.08－2016.12 California State University, Fresno 金融 本科

2022.09－2026.06 浙江大学 管理科学与工程 硕士